RESTORING THE RIGHT MEASURE: POSTERIOR-CONSISTENT FLOW MATCHING FOR PDE INVERSE PROBLEMS

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ABSTRACT

Flow matching models have emerged as a powerful paradigm for partial differential equation (PDE) inverse problems and uncertainty quantification (UQ). However, existing zero-shot hard-constrained projection samplers fail to preserve the true Bayesian conditional posterior, inducing posterior mode displacement and probability measure distortion over the constraint manifold. Conversely, exact alternatives either rely on inflexible retraining or expensive measure-corrected Markov chain Monte Carlo (MCMC) sampling. Recognizing a fundamental discrepancy in terminal targets between unconditional flows and constrained inverse problems, we propose the **Conditional Minimal Step-Wise Perturbation (CMSP)** framework. CMSP casts constrained generation as a conditional transport problem, constructing a Bayes-optimal conditional flow that redirects the trajectory's terminal attractor from the ambient data space toward the target conditional posterior. Furthermore, we introduce **Pre-Image Invariant Spectral Transfer (PIS)** to achieve efficient matrix-free velocity evaluation. Empirical evaluations across geometric posterior sweeps and high-dimensional physical PDEs demonstrate that CMSP achieves faithful posterior alignment and sets a state-of-the-art accuracy in uncertainty quantification among zero-shot hard-constrained methods.

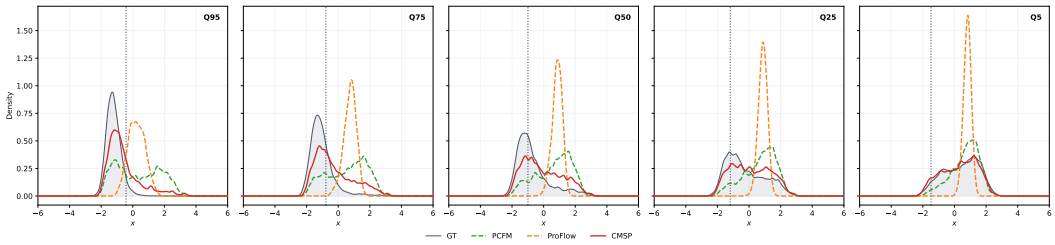


Figure 1: Multi-task posterior density profiles along the parabolic manifold across five posterior mass ratios. The vertical dotted line indicates the vertex center. While existing paradigms exhibit mode displacements and fail to adapt to varying geometric cut-offs, our proposed CMSP reconstructs both the dominant peaks and asymmetric tail geometries better.

1 INTRODUCTION

Continuous generative models, particularly Flow Matching (FM) (Lipman et al., 2023) and Functional Flow Matching (FFM) (Kerrigan et al., 2023), provide a powerful paradigm for physics-informed inverse problems in continuous functional spaces, such as reconstructing velocity fields from sparse sensors or identifying boundary parameters in partial differential equations (PDEs) (Stuart, 2010; Raissi et al., 2019).

Traditional paradigms incorporate physical information through penalties or gradient-based guidance (Bastek et al., 2025; Ben-Hamu et al., 2024). To enforce hard constraints without retraining, zero-shot inference-time methods rely on terminal projection schemes that extrapolate intermediate states

to candidate predictions and execute displacement projections onto $\mathcal{M}_1$ during integration (Cheng et al., 2025; Utkarsh et al., 2025; Yu et al., 2026). Furthermore, the Minimal Step-Wise Perturbation (MSP) framework rectifies surplus velocity arising from invalid linear interpolation assumptions after terminal projection (Anonymous, 2026).

Despite their success in achieving constraint feasibility, existing hard-projection paradigms exhibit a distributional limitation: satisfying physical constraints does not ensure that the generated samples follow the true Bayesian conditional posterior (Xu et al., 2026). In scientific inverse problems, feasible solutions to governing physical equations typically form a high-dimensional manifold containing diverse physical configurations. Existing hard-projection methods focus on eliminating physical residuals by moving samples onto this constraint manifold, but do not ensure that the posterior probability mass is correctly distributed across different regions of the manifold. Consequently, even when every generated sample satisfies the physical constraints, the ensemble may assign incorrect relative probabilities to competing feasible physical solutions, undermining the reliability of downstream uncertainty quantification.

Existing remedies for recovering the correct conditional posterior are forced to abandon zero-shot projection paradigms. On one hand, MCMC-based samplers explicitly incorporate the co-area correction (Xu et al., 2026), but their iterative manifold operations incur expensive computational costs. On the other hand, amortized conditional models learn measure-corrected posteriors directly from data (Zhou et al., 2026; Dasgupta et al., 2025); however, they require retraining dedicated networks for specific constraint settings. Consequently, obtaining a faithful conditional posterior while preserving both constraint satisfaction and zero-shot efficiency remains a critical challenge.

To address this challenge, the key observation underlying our framework is a fundamental discrepancy in terminal targets between unconditional flow matching and physics-constrained inverse problems. While a pretrained vanilla flow transports a source distribution p_0 toward the ambient target distribution p_1, physics-constrained inverse tasks require guiding probability paths toward a target posterior $\pi^\star$ defined strictly on the constraint manifold $\mathcal{M}_1 = \{u \mid \mathcal{C}(u) = \mathbf{0}\}$. To bridge this gap, we show that a conditional flow embedded with co-area measure geometry and posterior covariance guidance can be induced directly from the pretrained unconstrained flow. This redirects the trajectory's terminal attractor from the ambient distribution to the target conditional posterior, establishing a principled theoretical foundation for zero-shot constrained generation without retraining.

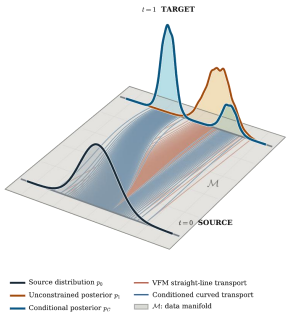


Figure 2: Mechanism of conditional transport.

Building upon this insight, we propose the **Conditional Minimal Step-Wise Perturbation (CMSP)** framework. CMSP leverages the Federer co-area disintegration formula to derive a Bayes-optimal conditional vector field directly from the pretrained unconstrained flow, thereby mitigating measure distortion and posterior variance collapse. To reduce the computational and memory costs associated with dense matrix inversion when evaluating the conditional velocity, we introduce **Pre-Image Invariant Spectral Transfer (PIS)** as an efficient, matrix-free computation scheme. Finally, CMSP executes discrete state updates governed by the variational principle of Minimal Step-Wise Perturbation (Anonymous, 2026), ensuring exact hard constraint satisfaction while maintaining dynamic consistency.

The main contributions of this paper are summarized as follows:

- **Theoretical Formulation:** We propose the **Conditional Minimal Step-Wise Perturbation (CMSP)** framework for PDE inverse problems. By addressing the discrepancy in terminal targets of unconditional flows, CMSP induces a Bayes-optimal conditional vector field that redirects the terminal attractor toward the true posterior, thereby mitigating posterior variance collapse.
- **Efficient Computation:** We introduce **Pre-Image Invariant Spectral Transfer (PIS)** for matrix-free evaluation of the covariance-weighted correction, requiring only $\mathcal{O}(d+m)$ working memory in the inner loop.
- **Empirical Validation:** We perform evaluations across geometric posterior sweeps and high-dimensional PDE inverse tasks. CMSP achieves exact constraint enforcement and better posterior

Table 1: Systematic comparison of generative methods for constrained inverse problems.

Methods	Exact Constraint	Dynamic Consistency	Co-Area Consistency	MCMC-Free
DiffusionPDE, D-Flow	✗	✗	✗	✓
ECI, PCFM, ProFlow	✓	✗	✗	✓
MSP	✓	✓	✗	✓
CoCoS	✓	✗	✓	✗
CMSP (Ours)	✓	✓	✓	✓

alignment, setting a new state-of-the-art in uncertainty quantification accuracy among zero-shot hard-constrained methods.

2 Related Works

2.1 Generative Modeling for PDE Inverse Problems

Learning-based PDE solvers, such as physics-informed neural networks and neural operators, approximate physical fields by embedding governing equations or learning operator mappings (Raissi et al., 2019; Li et al., 2021; Lu et al., 2021; Li et al., 2023). For ill-posed inverse problems with partial observations, generative models offer a probabilistic framework to represent multiple plausible solutions. Foundational diffusion and flow-based generative paradigms (Ho et al., 2020; Song et al., 2021; Lipman et al., 2023; Liu et al., 2022; Albergo et al., 2025) have been naturally extended to infinite-dimensional function spaces via Denoising Diffusion Operators and Functional Flow Matching (Lim et al., 2025; Kerrigan et al., 2023). Generative priors have been successfully applied to fluid systems, PDE forecasting, and partial-observation inference (Jacobsen et al., 2024; Lippe et al., 2023; Huang et al., 2024). To enforce physical plausibility, recent approaches incorporate physics-informed diffusion penalties (Bastek et al., 2025) or optimize pretrained generative flows via gradient guidance (Ben-Hamu et al., 2024). However, these soft-guidance methods cannot guarantee strict equality constraint satisfaction, highlighting the need for hard-constrained sampling paradigms.

2.2 Hard-Constrained Generative Sampling

Physical constraints can be incorporated into learning systems via training-time structural design or embedded optimization. Architectural approaches enforce conservation laws by constructing divergence-free networks or structural inductive biases (Hansen et al., 2024; Richter-Powell et al., 2022). Alternatively, optimization-based methods embed differentiable solvers or convex layers into training to enforce hard constraints (Donti et al., 2021; Agrawal et al., 2019; Négiar et al., 2023). Complementarily, inference-time methods seek to impose constraints directly on pretrained generative models without retraining.

To enforce constraints during sampling, trajectory-based methods like ECI (Cheng et al., 2025), PCFM (Utkarsh et al., 2025), and Projected Diffusion Models (Christopher et al., 2024) apply gradient-free updates, nonlinear corrections, or manifold projections. Expanding this framework, ProFlow incorporates terminal corrections into generative paths via proximal guidance (Yu et al., 2026), Gauge Flow Matching enforces convex constraints through gauge formulations (Li et al., 2026), and optimal-control-based models address terminally constrained generation (Gao et al., 2026). Together, these inference-time methods enable physically feasible sampling without retraining, motivating deeper investigation into whether strict constraint satisfaction truly recovers the correct conditional distribution.

2.3 Posterior Consistency in Constrained Generation

Conditioning probability distributions on equality-constrained manifolds requires careful treatment of the induced measure. Geometrically, the co-area formula establishes the measure-theoretic foundation for level sets (Federer, 1996), while the Fixman correction addresses geometric bias in constrained statistical mechanics (Fixman, 1974). Related works extend these principles to constrained Langevin

dynamics and manifold sampling (Lelievre et al., 2012), with the Borel–Kolmogorov paradox highlighting the subtleties of conditioning over measure-zero sets (Trésor & Lukashchuk, 2026).

These measure-theoretic subtleties directly impact Bayesian inference in physical systems, where satisfying governing equations does not guarantee sampling from the true posterior. While Bayesian PINNs (Yang et al., 2021), DGenNO (Zang & Koutsourelakis, 2025), and flow-matching Transformers (Sherki et al., 2025) provide probabilistic frameworks for PDE uncertainty quantification, recent works explicitly address consistency in constrained sampling. Furthermore, CoCoS demonstrates the necessity of co-area corrections to obtain posterior-consistent samples (Xu et al., 2026).

3 Background and Problem Setup

3.1 PDE Inverse Problems and Constraint Manifolds

Let $u \in \mathbb{R}^d$ denote a discretized PDE state. The prescribed physical and observational constraints define the feasible manifold $\mathcal{M}_1 \triangleq \{u \in \mathbb{R}^d : \mathcal{C}(u) = \mathbf{0}\}$, where $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ with $m \leq d$, and the corresponding constraint Jacobian $J(u) \in \mathbb{R}^{m\times d}$ has full row rank in a neighborhood of $\mathcal{M}_1$. Our goal is to construct the probability distribution of states conditioned on these constraints.

3.2 Vector-Field Prior

Let $t \in [0, 1]$ denote the flow time, and let $v^v(u, t) \in \mathbb{R}^d$ be a pretrained velocity field from Vanilla Flow Matching (VFM), transporting a source distribution p_0 to an unconstrained target distribution p_1. The vector field induces the flow map $\phi_t^v : \mathbb{R}^d \to \mathbb{R}^d$ via $d\phi_t^v(u_0)/dt = v^v(\phi_t^v(u_0), t)$ with $\phi_0^v(u_0) = u_0$. The pushforward measure $p_t^v \triangleq (\phi_t^v)_{\#}p_0$ satisfies the continuity equation $\partial_t p_t^v(u) + \nabla_u \cdot (p_t^v(u)v^v(u, t)) = 0$, with temporal boundaries $p_0^v = p_0$ and $p_1^v = p_1$. We treat VFM as a fixed generative prior $p_0 \xrightarrow{v^v} p_1$.

4 Conditional Minimal Step-Wise Perturbation

4.1 Inducing the Conditional Vector Field

In physical inverse problems, the terminal constraint $\mathcal{C}(u) = \mathbf{0}$ is typically underdetermined with $m < d$, which means infinitely many states satisfy the constraint. Simply finding an arbitrary feasible point on $\mathcal{M}_1$ is physically meaningless; our true objective is to capture which feasible states are physically probable, as characterized by the Bayesian conditional posterior $\pi^\star$ supported on $\mathcal{M}_1$. By the co-area formula and the disintegration theorem, this target measure integrates the unconstrained data density p_1 with the local constraint geometry:

$$d\pi^\star(u) = \frac{1}{Z} \frac{p_1(u)}{\sqrt{\det(J(u)J(u)^\top)}}\, d\mathcal{H}^{d-m}(u), \qquad u \in \mathcal{M}_1, \tag{1}$$

where $Z = \int_{\mathcal{M}_1} p_1(u)[\det\big(J(u)J(u)^\top\big)]^{-1/2} d\mathcal{H}^{d-m}(u)$ is the normalizing partition function.

While the unconstrained VFM transports the prior p_0 to the unconstrained target p_1, the constrained task requires a matching conditional vector field to transport p_0 to the posterior $\pi^\star$.

Definition 1 (Conditional VFM (CVFM))**.** A CVFM defines a velocity field $v^c : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d$ whose flow map ϕ_t^c transports the source distribution p_0 directly to the conditional posterior $\pi^\star$ via the continuity equation:

$$\partial_t p_t^c(u) + \nabla_u \cdot (p_t^c(u)v^c(u, t)) = 0, \qquad p_0^c = p_0, \quad p_1^c = \pi^\star, \tag{2}$$

where $p_t^c \triangleq (\phi_t^c)_{\#}p_0$. Along displacement interpolants connecting $X_0 \sim p_0$ to the constrained target $X_1 \sim \pi^\star$, the optimal velocity is governed by the conditional expectation $v^c(u, t) = (\mathbb{E}[X_1 \mid X_t = u, X_1 \in \mathcal{M}_1] - u)/(1 - t)$.

The fundamental distinction between VFM and CVFM lies in the terminal attractor of their probability trajectories: while v^v transports mass toward the unconstrained data prior p_1, v^c steers trajectories under posterior covariance guidance to concentrate strictly on the conditional distribution $\pi^\star$.

To construct v^c from the pretrained unconstrained field v^v without retraining, we query the unconstrained endpoint distribution $q^v(x_1 \mid u_t) \triangleq p(X_1 = x_1 \mid X_t = u_t)$ implied by v^v, restrict this distribution onto the constraint manifold $\mathcal{M}_1$ via the co-area formula to obtain the conditional posterior $q^c(x_1 \mid u_t)$, and finally compute its conditional mean μ_t^c.

Proposition 1 (Conditional Velocity Field Induced by the Prior Flow)**.** For any query state u_t at time $t \in [0, 1)$, the Bayes-optimal CVFM velocity field $v^c(u_t, t)$ in Definition 1 is given in closed form by:

$$v^c(u_t, t) = \frac{1}{1-t}\left[\frac{\displaystyle\int_{\mathcal{M}_1} x_1\, q^v(x_1 \mid u_t)\left[\det\big(J(x_1)J(x_1)^\top\big)\right]^{-1/2} d\mathcal{H}^{d-m}(x_1)}{\displaystyle\int_{\mathcal{M}_1} q^v(x_1 \mid u_t)\left[\det\big(J(x_1)J(x_1)^\top\big)\right]^{-1/2} d\mathcal{H}^{d-m}(x_1)} - u_t\right]. \quad (3)$$

Proposition 1 conditions the probability distribution before taking its mean, which guarantees that the resulting mean $\mu_t^c \triangleq \mathbb{E}_{q^c}[X_1 \mid X_t = u_t]$ is pulled toward regions with high posterior probability.

4.2 Theoretical Approximation

Evaluating the exact velocity Eq. 3 requires high-dimensional surface integration over the curved manifold $\mathcal{M}_1$, which is computationally intractable. To render this computation tractable, we perform a local approximation around the endpoint prediction μ_t^v: we model the endpoint posterior as a local Gaussian $q^v(x_1 \mid u_t) \approx \mathcal{N}(\mu_t^v, \Sigma_t^v)$, where the covariance Σ_t^v captures the directional uncertainty of the flow, and linearize the curved manifold by its tangent hyperplane through μ_t^v, expressed as $J_t(x_1 - \mu_t^v) + \mathcal{C}(\mu_t^v) = \mathbf{0}$ with the Jacobian $J_t \triangleq \nabla_u \mathcal{C}(\mu_t^v) \in \mathbb{R}^{m\times d}$.

These two local approximations convert the intractable manifold integration into a standard Gaussian conditioning problem on an affine subspace. Because the constraint is linearized, the co-area factor $\det\big(J_t J_t^\top\big)$ is constant across the tangent plane and cancels out, yielding a closed-form solution.

Proposition 2 (Local Tractable Conditional Velocity Field)**.** Under the local Gaussian and linearized constraint approximations, the CVFM velocity field admits the closed-form expression:

$$\boxed{\widehat{v}^c(u_t, t) = v^v(u_t, t) - \frac{1}{1-t}\Sigma_t^v J_t^\top \left(J_t \Sigma_t^v J_t^\top\right)^{-1} \mathcal{C}(\mu_t^v),} \quad (4)$$

where the local endpoint covariance $\Sigma_t^v \triangleq \mathrm{Cov}_{q^v}(X_1 \mid X_t = u_t)$ is induced by the spatial Jacobian of the flow field:

$$\Sigma_t^v = \frac{(1-t)^2}{t}\left[I_d + (1-t)\nabla_u v^v(u_t, t)\right]\left[I_d + (1-t)\nabla_u v^v(u_t, t)\right]^\top. \quad (5)$$

The operator $\Sigma_t^v J_t^\top (J_t \Sigma_t^v J_t^\top)^{-1}$ in Eq. 4 is an oblique projection under the Mahalanobis metric. It corrects constraint violations along directions where the flow exhibits highest uncertainty and carries the most probability mass. This ensures the correction path respects the natural data correlations learned by the prior instead of artificially projecting to the nearest geometric point.

4.3 Pre-image Invariant Spectral Transfer

While Eq. 4 provides an analytical formula, direct evaluation requires forming and inverting the dense $m \times m$ matrix $J_t \Sigma_t^v J_t^\top$, with $\mathcal{O}(m^3)$ factorization cost and substantial memory overhead from dense intermediate tensors. For fine-grid PDE problems with large m, these costs can make direct evaluation impractical. To avoid explicitly materializing this flow-weighted Gram matrix, we shift from static terminal algebra to flow dynamics, pulling the constraint backward along the flow.

We define the dynamic pre-image manifold at current time t:

$$\mathcal{M}_t \triangleq F_t^{-1}(\mathcal{M}_1) = \{u \in \mathbb{R}^d \mid \mathcal{C}(F_t(u)) = \mathbf{0}\}, \quad (6)$$

where $F_t : u_t \mapsto \mu_t^v$ denotes the local endpoint predictor induced by the VFM field. Physically, $\mathcal{M}_t$ represents the zero-intervention set: any state on $\mathcal{M}_t$ naturally flows into the feasible manifold $\mathcal{M}_1$ under the unconstrained vector field without requiring any correction. Trajectories only violate constraints if they deviate from $\mathcal{M}_t$ in its normal directions. Let $P_t, Q_t \in \mathbb{R}^{d\times d}$ denote the tangent and normal projectors of $\mathcal{M}_t$ at u_t, and let $P_1, Q_1 \in \mathbb{R}^{d\times d}$ denote those of $\mathcal{M}_1$ at μ_t^v.

Proposition 3 (Pre-image Invariance)**.** The flow differential Φ_t strictly preserves tangency between the pre-image and terminal manifolds, satisfying:

$$Q_1 \Phi_t P_t \equiv \mathbf{0}. \tag{7}$$

This vanishing property provides a dimension reduction: moving along the tangent directions of $\mathcal{M}_t$ only shifts the terminal state within the constraint manifold without violating the physics. Constraint violation is solely caused by perturbations in the normal direction. Consequently, the correction problem decouples from the full d-dimensional space and is isolated strictly to the m-dimensional normal subspace.

This one-way property ensures that the differential Φ_t is block upper-triangular between the pre-image and terminal frames. Under the flow-induced covariance factorization $\Sigma_t^v \propto \Phi_t \Phi_t^\top$, where $\Phi_t = I_d + (1-t)\nabla_u v^v(u_t, t)$, this structure triggers an exact algebraic cancellation in the Schur complement:

$$P_1(\Phi_t \Phi_t^\top) Q_1 \left[Q_1 (\Phi_t \Phi_t^\top) Q_1 \right]^{-1} \equiv B_t C_t^{-1}, \tag{8}$$

where $B_t \triangleq P_1 \Phi_t Q_t$ maps normal perturbations to terminal tangent displacements, and $C_t \triangleq Q_1 \Phi_t Q_t$ governs the normal-to-normal error propagation.

Remarkably, the dense quadratic matrix inversion $(C_t C_t^\top)^{-1}$ cancels out entirely, reducing the coupling to the linear operator transfer $B_t C_t^{-1}$. Physically, this operator acts as a two-stage sequential transfer: C_t^{-1} first determines the current normal nudge that cancels the terminal constraint defect $r_t \triangleq J_t^\top (J_t J_t^\top)^{-1} \mathcal{C}(\mu_t^v)$, and B_t then accounts for the tangent displacement induced by this normal correction through the flow. Computing the required normal pre-image $q^* = C_t^{-1} r_t$ is equivalent to minimizing the defect energy $\min_{q \in N_t} \frac{1}{2} \|C_t q - r_t\|^2$.

Proposition 4 (PIS Conditional Velocity Field)**.** Let $q_{t,K} \approx C_t^{-1} r_t$ be the approximate pre-image normal perturbation. The PIS-approximated CVFM velocity field $\widetilde{v}^c(u_t, t)$ is given by:

$$\boxed{\widetilde{v}^c(u_t, t) = \frac{z_t^C - u_t}{1-t}, \qquad z_t^C = \Pi_{\mathcal{M}_1} \left(\mu_t^v - r_t - B_t q_{t,K} \right).} \tag{9}$$

In practice, we evaluate $q_{t,K}$ through K steps of exact line-search descent on $\min_{q \in \mathcal{N}_t} \frac{1}{2} \|C_t q - r_t\|^2$, which treats the linear inverse as a strictly monotonic energy-dissipation process in the normal space. Starting from $q_0 = \mathbf{0}$ with initial residual $e_0 = -r_t$, each iteration executes:

$$g_k = C_t^\top e_k, \qquad \gamma_k = \frac{\|g_k\|^2}{\|C_t g_k\|^2}, \qquad q_{k+1} = q_k - \gamma_k g_k, \qquad e_{k+1} = e_k - \gamma_k C_t g_k. \tag{10}$$

Each PIS iteration requires one Vector-Jacobian Product (VJP) for $C_t^\top$ and one Jacobian-Vector Product (JVP) for C_t, reusing the flow network's forward intermediate activations. Outer-step operations are performed once per flow step, outside the K-iteration loop, so increasing K adds only the cost of the PIS iterations.

Peak Memory Reduction via PIS. Direct evaluation of the velocity correction in Eq. 4 requires forming and inverting the flow-weighted Gram matrix $J_t \Sigma_t^v J_t^\top \in \mathbb{R}^{m \times m}$. Since $\Sigma_t^v \propto \Phi_t \Phi_t^\top$, where $\Phi_t = I_d + (1-t)\nabla_u v^v \in \mathbb{R}^{d \times d}$, this computation incurs substantial memory overhead from dense intermediate tensors and matrix-factorization workspace, potentially causing out-of-memory (OOM) failures in fine-grid PDE discretizations. PIS exploits pre-image invariance to decompose the covariance-weighted correction into the normal defect $r_t = J_t^\top (J_t J_t^\top)^{-1} \mathcal{C}(\mu_t^v)$ and the tangent-transfer term $B_t C_t^{-1} r_t$. It approximates $C_t^{-1} r_t$ through matrix-free operations, avoiding explicit materialization of $J_t \Sigma_t^v J_t^\top$ and the dense flow Jacobian Φ_t. The inner loop maintains only a fixed number of state- and constraint-space vectors (q, e, g, w), substantially reducing memory overhead.

4.4 Update Framework

Having constructed $\widetilde{v}^c(u_t, t)$, we now establish the discrete temporal update rule. We adopt the variational principle of Minimum Step-Wise Perturbation (MSP) to advance the CVFM dynamics. The central objective is to determine the subsequent state $u_{t+\Delta t}$ that minimizes the local perturbation relative to the unconstrained step $u_t + \Delta t\, \widetilde{v}^c(u_t, t)$, while ensuring that the forward-propagated trajectory satisfies the terminal constraint $\mathcal{C}(u_1) = \mathbf{0}$. Let $u_0 \sim p_0$ be the initial sample at $t = 0$.

Table 2: Uncertainty quality and benchmark on the $d = 8$ Darcy inverse problem. Evaluated against the i.i.d. rejection arbiter (Gold Floor).

Method	$\overline{W}_1$	$\times$floor	mean err	std rel. err	cov_{90}	Time (s)
Gold (Floor)	*0.0117*	*1.0×*	—	—	*0.90*	—
CoCoS	0.0583	5.0×	0.1799	0.034	0.89	227.1
Guided Langevin	0.1211	10.3×	0.3537	0.338	0.93	268.1
Soft-penalty ($\sigma = 0.02$)	0.0854	7.3×	0.3148	0.116	0.92	0.2
PCFM	0.1829	15.6×	0.2578	0.547	0.96	2.1
CMSP (Ours)	**0.0573**	**4.9×**	**0.1269**	**0.106**	**0.91**	**6.5**

Proposition 5 (Conditional MSP Update)**.** The discrete state update $u_{t+\Delta t}$ that minimizes the step-wise state perturbation subject to terminal constraint satisfaction along the optimal transport ray from u_0 is given by:

$$\boxed{u_{t+\Delta t} = (1 - t - \Delta t)u_0 + (t + \Delta t)u_1^{\text{corr}},} \tag{11}$$

where $u_1^{\text{corr}} \triangleq \Pi_{\mathcal{M}_1}(u_1^{\text{ref}})$ is the Euclidean projection onto $\mathcal{M}_1$ of the equivalent reference endpoint:

$$u_1^{\text{ref}} \triangleq \frac{u_t + \Delta t\, \widetilde{v}^c(u_t, t) - (1 - t - \Delta t)u_0}{t + \Delta t}. \tag{12}$$

The execution of Proposition 5 proceeds in four distinct steps: (i) compute the conditional velocity $\widetilde{v}^c(u_t, t)$; (ii) construct the reference endpoint u_1^{ref} in Eq. 12; (iii) project this endpoint candidate onto the terminal constraint manifold to obtain u_1^{corr}; (iv) pull back from the initial sample u_0 along the corrected optimal transport ray to the next time level $t + \Delta t$. (see Algorithm 1 for the specific details and Appendix A for proofs of the theory.).

5 Experiments

5.1 Uncertainty Quality Benchmark across Paradigms.

We evaluate our framework on a $d = 8$ nonlinear Darcy flow inverse problem with $m = 3$ sparse internal pressure observations under the hard constraint $C(\theta) = \mathcal{H}(\mathcal{G}(\theta)) - y_{\text{obs}} = \mathbf{0}$. To establish an i.i.d. gold-standard reference posterior, we collect $40{,}000$ rejection samples under a strict residual tolerance $\varepsilon = 0.006$.

This experiment provides a cross-paradigm characterization for broad uncertainty calibration. We compare CMSP against representative baselines spanning four distinct paradigms: (i) **Manifold MCMC** (CoCoS); (ii) **Inference-Time Guidance** (Guided Langevin, representing D-Flow / DiffusionPDE style); (iii) **Soft-Penalty Posterior** (PBFM/PIDM style filtering); and (iv) **Hard Projection** (PCFM). Full dataset construction and sampling-noise floor estimation details, together with exact hyperparameter configurations and step sizes are deferred to Appendix C.

All experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU (24 GB GDDR6X VRAM) and an Intel(R) Xeon(R) Platinum 8358P CPU (16 vCPUs @ 2.60 GHz).

Algorithm 1 CMSP Update

Require: $v^v, \mathcal{C}(u), u_0 \sim p_0, N, K$.
Ensure: u_1 with $\mathcal{C}(u_1) = \mathbf{0}$.

$\Delta t \leftarrow 1/N, \quad u \leftarrow u_0$
for $k = 0, \ldots, N-1$ **do**
  $t \leftarrow k \cdot \Delta t$
  $\mu_t^v \leftarrow u + (1-t)v^v(u,t)$
  $J_t \leftarrow \nabla_u \mathcal{C}(\mu_t^v)$
  $r_t \leftarrow J_t^\top (J_t J_t^\top)^{-1} \mathcal{C}(\mu_t^v)$
  $q \leftarrow \mathbf{0}, \quad e \leftarrow -r_t$
  for $j = 0, \ldots, K-1$ **do**
    $g \leftarrow C_t^\top e, \quad w \leftarrow C_t g$
    $\gamma \leftarrow \|g\|^2 / \|w\|^2$
    $q \leftarrow q - \gamma g, \quad e \leftarrow e - \gamma w$
  end for
  $z_t^C \leftarrow \Pi_{\mathcal{M}_1}(\mu_t^v - r_t - B_t q)$
  $\widetilde{v}^c \leftarrow (z_t^C - u)/(1-t)$
  $u_1^{\text{ref}} \leftarrow \frac{u + \Delta t\, \widetilde{v}^c - (1-t-\Delta t)u_0}{t+\Delta t}$
  $u_1^{\text{corr}} \leftarrow \Pi_{\mathcal{M}_1}(u_1^{\text{ref}})$
  $u \leftarrow (1 - t - \Delta t)u_0 + (t + \Delta t)u_1^{\text{corr}}$
end for
return u

Table 3: Quantitative benchmark on the translated parabola manifold across five posterior mass ratios. **Bold gold** indicates the best performance, and **standard bold** indicates the second. Full benchmark across all four manifolds is deferred to Appendix Table 6.

Method	Mean Field MSE (MMSE) ↓					Standard Deviation Field MSE (SMSE) ↓					Wasserstein-1 Distance (W_1) ↓				
	Q95	Q75	Q50	Q25	Q5	Q95	Q75	Q50	Q25	Q5	Q95	Q75	Q50	Q25	Q5
PCFM	2.498	2.986	2.264	1.022	0.083	11.581	5.364	1.889	0.569	0.056	1.546	1.630	1.369	0.925	0.305
ECI (nonlinear)	2.191	**2.772**	**2.144**	**0.976**	**0.079**	**10.089**	**4.941**	**1.771**	**0.537**	**0.052**	**1.443**	**1.564**	**1.325**	**0.898**	**0.287**
ProFlow	**2.113**	3.410	2.765	1.340	0.165	10.290	6.562	2.480	0.816	0.129	1.457	1.820	1.650	1.281	0.779
MSP	3.358	3.495	2.535	1.123	0.090	15.751	6.370	2.150	0.639	0.064	1.802	1.777	1.463	0.982	0.345
CMSP (**Ours**)	**0.389**	**0.672**	**0.515**	**0.138**	**0.015**	**1.877**	**1.249**	**0.439**	**0.078**	**0.009**	**0.624**	**0.785**	**0.650**	**0.349**	**0.096**

As evaluated in Table 2, the results demonstrate that CMSP achieves a distinct advantage in balancing statistical fidelity with computational scalability. The posterior accuracy of CMSP is comparable to the manifold Markov Chain Monte Carlo method CoCoS, achieving a mean error of 0.1269 and an average Wasserstein distance $\overline{W}_1$ of 0.0573. Simultaneously, it bypasses the performance bottleneck of iterative chains, achieving an approximately $35\times$ speedup (6.5 s vs. 227.1 s). Within the zero-shot hard-projection regime, CMSP effectively mitigates the variance collapse inherent in existing trajectory-projection solvers. Compared with PCFM, the marginal Wasserstein distance is reduced by over 68%.

5.2 Posterior Mass-Ratio Sweep Benchmark

To rigorously assess whether hard-constrained generative models preserve posterior consistency under severe geometric and measure-theoretic distortions, we benchmark on a nonlinearly 2D system. Specifically, we enforce the hard parabolic constraint $C(u) = u_2 - \left[0.2(u_1 - c)^2 + 0.8\right] = 0$. The generative flow is tasked with transporting a Gaussian source distribution $p_0 = \mathcal{N}(\mathbf{0}, \mathbf{I})$ to an anisotropic terminal ambient target $p_1 = \mathcal{N}\left(\begin{bmatrix}0.2\\2.5\end{bmatrix}, \begin{bmatrix}0.45 & 0.28\\0.28 & 0.35\end{bmatrix}\right)$.

By varying the vertex center $c \in \mathbb{R}$, we construct five distinct tasks (Q95, Q75, Q50, Q25, Q5) where the exact ground-truth conditional probability mass situated to the left of the vertex ($u_1 < c$) targets $95\%, 75\%, 50\%, 25\%$, and 5%, respectively. Given the complexity of the manifolds, we implement ECI's correction step using the same nonlinear projection solver as PCFM and denote this adapted variant as ECI (nonlinear). Quantitative results are summarized in Table 3. Fig. 1 shows the fine-grained posterior density profiles along the manifold.

Across all five mass-ratio configurations, CMSP consistently achieves superior fidelity, outperforming all baselines by up to an order of magnitude on both moment-level accuracy and overall distributional alignment. Specifically, relative to the strongest non-CMSP baseline for each setting, CMSP reduces MMSE by 75.8%–85.9% and SMSE by 74.7%–85.5%, alongside substantial reductions in W_1 distance. Strikingly, while unadjusted projection paradigms suffer from severe mode displacement and variance collapse, CMSP approximates the true Bayesian posterior. As visualized in Fig. 1, CMSP accurately mirrors the ground-truth distribution, closely reproducing both dominant modes and asymmetric tail geometries across all mass-ratio sweeps.

Furthermore, to evaluate robustness under diverse topological and geometric tasks, we extend our benchmark across three challenging regimes with different difficulties. Across all four distinct geometric topologies, CMSP achieves the best performance in 51 out of 60 evaluations, confirming that our conditional transport reliably prevents mode collapse even under severe curvature and topological discontinuities. Additional details, comprehensive evaluations on the other 2D geometries, and extended visualizations are provided in Appendix D.

5.3 High-Dimensional PDE Experiments

To evaluate the quality of the distributions generated by different zero-shot hard-projection methods, we compare the pointwise statistics of the generated solutions against those of the reference solutions. To mitigate sampling variability, we use 512 generated samples for each method on every task. We evaluate CMSP on four constrained generation tasks: the heat equation, reaction–diffusion initial conditions (IC), Burgers boundary conditions (BC), and Burgers initial conditions (IC). The quantitative results are reported in Table 4, and comparison of solutions on Heat Equation is provided

Table 4: Quantitative benchmark on high-dimensional PDE inverse problems across four physical systems. Bold font indicates the best performance for each metric (lower is better).

Dataset	Metric	CMSP (Ours)	MSP	ECI	PCFM	ProFlow
Heat Equation	MMSE / 10^{-2}	**0.008**	1.989	8.129	1.262	0.709
	SMSE / 10^{-2}	**0.009**	0.574	9.969	1.499	0.016
	FPD	**0.679**	10.213	12.084	6.476	4.409
Reaction-Diffusion IC	MMSE / 10^{-2}	**0.013**	0.381	1.026	0.165	0.357
	SMSE / 10^{-2}	**0.039**	1.877	9.041	0.579	0.049
	FPD	**0.571**	6.548	10.046	2.868	2.218
Burgers IC	MMSE / 10^{-2}	**0.274**	2.885	10.278	7.564	6.921
	SMSE / 10^{-2}	**0.533**	3.174	7.646	1.901	1.566
	FPD	**1.544**	7.309	7.598	6.851	6.155
Burgers BC	MMSE / 10^{-2}	**0.051**	0.604	1.038	0.908	1.070
	SMSE / 10^{-2}	**0.084**	0.216	0.155	0.161	0.278
	FPD	**1.543**	2.827	2.757	2.397	2.689

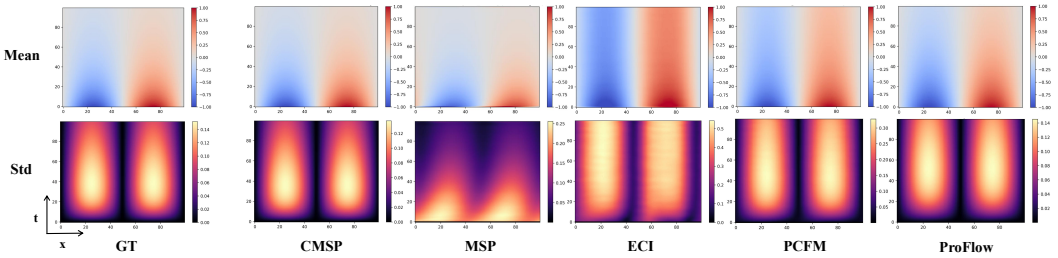


Figure 3: Comparison of generated solutions on the Linear Heat Equation task.

in Fig. 3. Detailed descriptions of the PDE systems, constraint specifications, and experimental parameter settings, together with more visualization are provided in Appendix E. CMSP consistently achieves the best performance across all four PDE tasks, attaining the lowest MMSE, SMSE, and FPD in every setting. The results demonstrate that CMSP accurately captures both pointwise statistics and the overall distribution of constrained solutions.

By plotting the Pareto frontier (see Fig. 4), we further evaluate the trade-off between MMSE and computational efficiency with 256 generated samples for each method on the Burgers BC task. CMSP with $K = 1$ achieves an MMSE significantly lower than other peer methods, while its computational cost remains within an acceptable range. Increasing sampling steps, scaling up mixing iterations (ECI), or introducing guidance mechanisms (PCFM, penalty weight $\lambda = 1.0$, step size $= 0.01$, num steps $= 20$) all fail to eliminate the performance gap caused by the lack of co-area information.

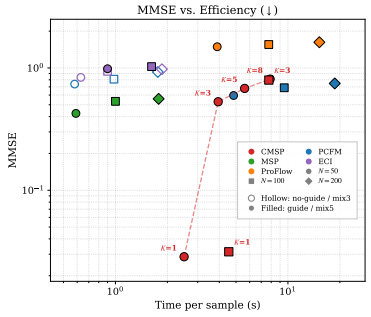


Figure 4: Pareto frontier.

Another finding is that CMSP achieves its best at $K = 1$, whereas increasing K leads to a degradation. A plausible explanation is that PIS iterations exhibit an implicit spectral filtering property: a single-step iteration preferentially captures dominant, well-conditioned components in the residual, while suppressing error amplification along ill-conditioned singular directions. This phenomenon is not universally present. Our experiments across other tasks indicate that increasing K does not always lead to performance drops; for the remaining tasks, typically only a small number of inner iterations, namely $K \leq 3$, are sufficient to approach the optimal performance reachable by CMSP under the corresponding sampling step budget. The above results further validate the high efficiency of CMSP.

6 CONCLUSION

In this paper, we proposed the **Conditional Minimal Step-Wise Perturbation (CMSP)** framework to address the limitation of zero-shot hard-constrained methods in neglecting true Bayesian posterior sampling. By incorporating the Federer co-area disintegration formula into flow dynamics, CMSP induces a Bayes-consistent conditional vector field from an unconstrained flow, substantially mitigating the posterior variance collapse and mode displacement inherent in standard geometric projections. Extensive experiments demonstrate that CMSP achieves superior posterior alignment and reliable uncertainty quantification. Looking forward, we plan to extend CMSP to complex real-world scenarios and higher-dimensional physical inverse problems in continuous functional spaces.

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A Proofs of Theoretical Results

A.1 Proof of Proposition 1

Proof. Let $u_t \in \mathbb{R}^d$ be the query state at continuous flow time $t \in [0, 1)$. Under the linear displacement path adopted by the flow matching framework, the time-dependent state is defined by the linear interpolation

$$X_t = (1 - t)X_0 + tX_1. \tag{13}$$

Differentiating Eq. 13 with respect to t gives the instantaneous path velocity $\dot{X}_t = X_1 - X_0$. Expressing the initial state X_0 in terms of X_t and X_1 yields the algebraic identity

$$\dot{X}_t = X_1 - X_0 = \frac{X_1 - X_t}{1 - t}. \tag{14}$$

By Definition 1, the conditional velocity field $v^c(u_t, t)$ is defined as the conditional expectation of the path velocity conditioned on the event that the flow reaches the state $X_t = u_t$ and terminates on the constraint manifold $X_1 \in \mathcal{M}_1$:

$$v^c(u_t, t) = \mathbb{E}\left[\dot{X}_t \;\middle|\; X_t = u_t, X_1 \in \mathcal{M}_1\right]. \tag{15}$$

Substituting the identity Eq. 14 into Eq. 15 and factoring out the deterministic evaluation point u_t alongside the scalar prefactor $(1 - t)^{-1}$, we obtain

$$v^c(u_t, t) = \mathbb{E}\left[\frac{X_1 - u_t}{1 - t} \;\middle|\; X_t = u_t, X_1 \in \mathcal{M}_1\right] = \frac{\mu_t^c - u_t}{1 - t}, \tag{16}$$

where $\mu_t^c \triangleq \mathbb{E}\left[X_1 \mid X_t = u_t, X_1 \in \mathcal{M}_1\right]$ denotes the conditional centroid of the terminal state constrained to $\mathcal{M}_1$.

To evaluate μ_t^c, let $q^v(x_1 \mid u_t) \triangleq p(X_1 = x_1 \mid X_t = u_t)$ denote the unconstrained endpoint posterior density. Because the terminal constraint manifold $\mathcal{M}_1 = \{x_1 \in \mathbb{R}^d \mid \mathcal{C}(x_1) = \mathbf{0}\}$ has codimension $m \geq 1$ in $\mathbb{R}^d$, it constitutes a set of Lebesgue measure zero. Assuming that the constraint mapping $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ is continuously differentiable with its Jacobian $J(x_1) \triangleq \nabla_u \mathcal{C}(x_1) \in \mathbb{R}^{m \times d}$ having full row rank in a neighborhood of $\mathcal{M}_1$, we apply the Federer co-area formula. For any bounded Borel test function $\psi \in C_b(\mathbb{R}^d)$, the integral of the posterior distribution factors into an iterated integral over the level sets $\mathcal{C}^{-1}(y)$:

$$\int_{\mathbb{R}^d} \psi(x_1) q^v(x_1 \mid u_t)\, dx_1 = \int_{\mathbb{R}^m} \left(\int_{\mathcal{C}^{-1}(y)} \frac{\psi(x_1) q^v(x_1 \mid u_t)}{\sqrt{\det(J(x_1)J(x_1)^\top)}}\, d\mathcal{H}^{d-m}(x_1) \right) dy. \tag{17}$$

The co-area disintegration therefore yields the following canonical conditional measure on $\mathcal{M}_1 = \mathcal{C}^{-1}(\mathbf{0})$:

$$dq^c(x_1 \mid u_t) = \frac{1}{Z_t} \frac{q^v(x_1 \mid u_t)}{\sqrt{\det(J(x_1)J(x_1)^\top)}}\, d\mathcal{H}^{d-m}(x_1), \qquad x_1 \in \mathcal{M}_1, \tag{18}$$

where the normalizing partition function

$$Z_t \triangleq \int_{\mathcal{M}_1} \frac{q^v(x_1 \mid u_t)}{\sqrt{\det(J(x_1)J(x_1)^\top)}}\, d\mathcal{H}^{d-m}(x_1) \tag{19}$$

is assumed to satisfy $0 < Z_t < \infty$ so that $q^c(\cdot \mid u_t)$ is a well-defined probability measure. Under this conditional measure, the conditional centroid μ_t^c is expressed as the first moment

$$\mu_t^c = \int_{\mathcal{M}_1} x_1\, dq^c(x_1 \mid u_t) = \frac{\displaystyle\int_{\mathcal{M}_1} x_1\, q^v(x_1 \mid u_t) \left[\det\big(J(x_1)J(x_1)^\top\big)\right]^{-1/2} d\mathcal{H}^{d-m}(x_1)}{\displaystyle\int_{\mathcal{M}_1} q^v(x_1 \mid u_t) \left[\det\big(J(x_1)J(x_1)^\top\big)\right]^{-1/2} d\mathcal{H}^{d-m}(x_1)}. \tag{20}$$

Substituting Eq. 20 into Eq. 16 directly yields the exact integral representation in Eq. 3, completing the proof. □

A.2 PROOF OF PROPOSITION 2

Proof. Let $u_t \in \mathbb{R}^d$ be the query state at flow time $t \in (0,1)$. Recall the unconstrained endpoint predictor $\mu_t^v \triangleq u_t + (1-t)v^v(u_t,t)$ and the exact integral representation of the conditional velocity field $v^c(u_t,t) = (\mu_t^c - u_t)/(1-t)$ established in Proposition 1, where μ_t^c denotes the conditional centroid over the nonlinear manifold $\mathcal{M}_1$. To obtain a computationally tractable formulation that bypasses high-dimensional surface integration, we construct a local Gaussian-affine approximation around μ_t^v and solve the resulting conditional distribution analytically.

We first formulate the local Gaussian approximation of the unconstrained endpoint posterior $q^v(x_1 \mid u_t) \triangleq p(X_1 = x_1 \mid X_t = u_t)$. Rather than attempting an intractable global Hessian inversion of an arbitrary data prior, we adopt the flow-induced covariance factorization:

$$\Sigma_t^v = \frac{(1-t)^2}{t}\Phi_t\Phi_t^\top, \qquad \Phi_t \triangleq \nabla_u \mu_t^v = I_d + (1-t)\nabla_u v^v(u_t,t) \in \mathbb{R}^{d\times d}. \tag{21}$$

This representation directly models the pushforward of the isotropic reference uncertainty under the linearized endpoint mapping $F_t : u_t \mapsto \mu_t^v$. Under the assumption that the spatial flow Jacobian Φ_t is nonsingular for $t \in (0,1)$, the matrix Σ_t^v is strictly symmetric and positive-definite ($\Sigma_t^v \succ 0$). Approximating the endpoint distribution by the local Gaussian density centered at μ_t^v yields:

$$q^v(x_1 \mid u_t) \approx \mathcal{N}\left(x_1;\ \mu_t^v,\ \Sigma_t^v\right) \propto \exp\left(-\frac{1}{2}(x_1-\mu_t^v)^\top(\Sigma_t^v)^{-1}(x_1-\mu_t^v)\right). \tag{22}$$

Next, assuming that the constraint mapping $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ is of class C^2 in an open neighborhood of μ_t^v, we perform a first-order Taylor expansion:

$$\mathcal{C}(x_1) = \mathcal{C}(\mu_t^v) + J_t(x_1-\mu_t^v) + \mathcal{R}_{\mathcal{C}}(x_1), \tag{23}$$

where $J_t \triangleq \nabla_u \mathcal{C}(\mu_t^v) \in \mathbb{R}^{m\times d}$ is the constraint Jacobian evaluated at μ_t^v, and the remainder satisfies $\|\mathcal{R}_{\mathcal{C}}(x_1)\| \le \frac{1}{2}\sup_\xi \|\nabla_u^2 \mathcal{C}(\xi)\|\|x_1-\mu_t^v\|^2$. Neglecting this second-order curvature remainder defines the first-order affine approximation of the constraint manifold $\mathcal{M}_1$ around μ_t^v:

$$\mathcal{M}_{\text{lin}} \triangleq \left\{x_1 \in \mathbb{R}^d \;\middle|\; J_t(x_1-\mu_t^v) + \mathcal{C}(\mu_t^v) = \mathbf{0}\right\}. \tag{24}$$

On the affine subspace $\mathcal{M}_{\text{lin}}$, the constraint gradient $\nabla_{x_1}[J_t(x_1-\mu_t^v)+\mathcal{C}(\mu_t^v)] \equiv J_t$ is spatially invariant. Consequently, the co-area volume distortion term $[\det(J(x_1)J(x_1)^\top)]^{-1/2} = [\det(J_tJ_t^\top)]^{-1/2}$ is strictly constant across $\mathcal{M}_{\text{lin}}$, and thus cancels out identically between the numerator and denominator integrals of the conditional centroid representation in Eq. 3. As a result, the approximated conditional posterior measure $q^c(x_1 \mid u_t)$ reduces to the Gaussian distribution $\mathcal{N}(\mu_t^v, \Sigma_t^v)$ conditioned on the affine linear constraint in Eq. 24.

Because a Gaussian distribution restricted to an affine subspace remains Gaussian, its conditional mean $\widehat{\mu}_t^c \triangleq \mathbb{E}_{q^c}[X_1 \mid X_t = u_t]$ coincides with its mode on the affine constraint set $\mathcal{M}_{\text{lin}}$. This state is the unique solution to the linearly constrained quadratic optimization problem:

$$\widehat{\mu}_t^c = \arg\min_{x_1\in\mathbb{R}^d} \frac{1}{2}(x_1-\mu_t^v)^\top(\Sigma_t^v)^{-1}(x_1-\mu_t^v) \quad \text{s.t.} \quad J_t(x_1-\mu_t^v) + \mathcal{C}(\mu_t^v) = \mathbf{0}. \tag{25}$$

Introducing the Lagrange multiplier vector $\lambda \in \mathbb{R}^m$, the Lagrangian function associated with Eq. 25 is given by:

$$\mathcal{L}(x_1,\lambda) = \frac{1}{2}(x_1-\mu_t^v)^\top(\Sigma_t^v)^{-1}(x_1-\mu_t^v) + \lambda^\top\left[J_t(x_1-\mu_t^v) + \mathcal{C}(\mu_t^v)\right]. \tag{26}$$

Setting the gradient with respect to x_1 to zero yields the stationarity condition:

$$\nabla_{x_1}\mathcal{L} = (\Sigma_t^v)^{-1}(x_1-\mu_t^v) + J_t^\top\lambda = \mathbf{0} \implies x_1 - \mu_t^v = -\Sigma_t^v J_t^\top\lambda. \tag{27}$$

Multiplying Eq. 27 from the left by J_t and enforcing the affine feasibility condition $J_t(x_1-\mu_t^v) = -\mathcal{C}(\mu_t^v)$, we obtain:

$$-J_t\Sigma_t^v J_t^\top\lambda = -\mathcal{C}(\mu_t^v) \implies \lambda = \left(J_t\Sigma_t^v J_t^\top\right)^{-1}\mathcal{C}(\mu_t^v), \tag{28}$$

where the invertibility of $J_t\Sigma_t^v J_t^\top$ is guaranteed by the full row rank of J_t and the strict positive-definiteness of Σ_t^v. Substituting λ back into Eq. 27 gives the analytical conditional centroid:

$$\widehat{\mu}_t^c = \mu_t^v - \Sigma_t^v J_t^\top \left(J_t\Sigma_t^v J_t^\top\right)^{-1} \mathcal{C}(\mu_t^v). \tag{29}$$

Finally, substituting the conditional centroid $\widehat{\mu}_t^c$ into the velocity-displacement relation $v^c(u_t, t) = (\mu_t^c - u_t)/(1-t)$ and substituting $\mu_t^v = u_t + (1-t)v^v(u_t, t)$ yields:

$$\begin{aligned}
\widehat{v}^c(u_t, t) &= \frac{\widehat{\mu}_t^c - u_t}{1-t} \\
&= \frac{\mu_t^v - u_t}{1-t} - \frac{1}{1-t}\Sigma_t^v J_t^\top \left(J_t\Sigma_t^v J_t^\top\right)^{-1} \mathcal{C}(\mu_t^v) \\
&= v^v(u_t, t) - \frac{1}{1-t}\Sigma_t^v J_t^\top \left(J_t\Sigma_t^v J_t^\top\right)^{-1} \mathcal{C}(\mu_t^v),
\end{aligned} \tag{30}$$

which establishes Eq. 4 and completes the proof. □

A.3 Proof of Proposition 3

Proof. Let $u_t \in \mathcal{M}_t$, which by definition of the dynamic pre-image manifold $\mathcal{M}_t \triangleq F_t^{-1}(\mathcal{M}_1)$ implies that the forward endpoint prediction satisfies the terminal constraint $\mu_t^v = F_t(u_t) \in \mathcal{M}_1$. The proof proceeds by characterizing the tangent spaces of $\mathcal{M}_1$ and $\mathcal{M}_t$ as the null spaces of their respective constraint Jacobians and applying the multivariate chain rule.

By assumption, the constraint mapping $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ is continuously differentiable in an open neighborhood of $\mathcal{M}_1$, with its constraint Jacobian $J_t \triangleq \nabla_u \mathcal{C}(\mu_t^v) \in \mathbb{R}^{m\times d}$ having full row rank m. Furthermore, we assume that the spatial flow differential $\Phi_t \triangleq \nabla_u F_t(u_t) \in \mathbb{R}^{d\times d}$ is nonsingular. By Sylvester's rank inequality, the composite matrix $J_t\Phi_t \in \mathbb{R}^{m\times d}$ maintains full row rank $\operatorname{rank}(J_t\Phi_t) = m$, which guarantees that the composite mapping $\widetilde{\mathcal{C}}(u) \triangleq (\mathcal{C} \circ F_t)(u)$ is a submersion at u_t, and consequently the pre-image $\mathcal{M}_t$ is locally a smooth, embedded submanifold of dimension $d - m$.

By the regular value theorem, the tangent space of the terminal constraint manifold $\mathcal{M}_1$ at μ_t^v, denoted $T_1 \triangleq T_{\mu_t^v}\mathcal{M}_1$, is characterized as the kernel of the constraint Jacobian:

$$T_1 = \ker J_t = \left\{w \in \mathbb{R}^d \;\middle|\; J_t w = \mathbf{0}\right\}. \tag{31}$$

The terminal normal space $N_1 \triangleq T_1^\perp = \operatorname{range}(J_t^\top)$ is the orthogonal complement of T_1, and $Q_1 \in \mathbb{R}^{d\times d}$ denotes the orthogonal projector onto N_1. Consequently, for any vector $w \in \mathbb{R}^d$, the projection vanishes $Q_1 w = \mathbf{0}$ if and only if $w \in T_1$.

Next, we characterize the tangent space of the pre-image manifold $\mathcal{M}_t$ at u_t. Differentiating the composite constraint $\widetilde{\mathcal{C}}(u) = (\mathcal{C} \circ F_t)(u) = \mathbf{0}$ with respect to u at u_t using the multivariate chain rule yields:

$$\nabla_u \widetilde{\mathcal{C}}(u_t) = \nabla_u (\mathcal{C} \circ F_t)(u_t) = \nabla_u \mathcal{C}(F_t(u_t)) \cdot \nabla_u F_t(u_t) = J_t\Phi_t \in \mathbb{R}^{m\times d}. \tag{32}$$

The tangent space of the pre-image manifold at u_t, denoted $T_t \triangleq T_{u_t}\mathcal{M}_t$, is defined as the kernel of this composite Jacobian:

$$T_t = \ker(J_t\Phi_t) = \left\{\xi \in \mathbb{R}^d \;\middle|\; J_t\Phi_t\xi = \mathbf{0}\right\}. \tag{33}$$

Let $P_t \in \mathbb{R}^{d\times d}$ denote the orthogonal projector onto the pre-image tangent space T_t.

Now, consider an arbitrary vector $v \in \mathbb{R}^d$, and let $\xi \triangleq P_t v \in T_t$ be its tangent projection. By the kernel condition in Eq. 33, we have:

$$J_t\Phi_t\xi = J_t\left(\Phi_t P_t v\right) = \mathbf{0}. \tag{34}$$

Examining the transformed vector $w \triangleq \Phi_t P_t v \in \mathbb{R}^d$, Eq. 34 states that $J_t w = \mathbf{0}$, which by Eq. 31 establishes that $w \in \ker J_t = T_1$. This demonstrates that the flow differential Φ_t maps the pre-image tangent space into the terminal tangent space, i.e., $\Phi_t(T_t) \subseteq T_1$.

Because the terminal normal projector Q_1 annihilates any vector in T_1, applying Q_1 to $w = \Phi_t P_t v$ yields:

$$Q_1 \left(\Phi_t P_t v\right) = \mathbf{0}. \tag{35}$$

Since Eq. 35 holds identically for all $v \in \mathbb{R}^d$, the linear operator equality

$$Q_1 \Phi_t P_t \equiv \mathbf{0} \tag{36}$$

holds identically on $\mathbb{R}^d$, establishing Eq. 7 and completing the proof. □

A.4 Proof of Proposition 4

Proof. We begin by establishing the connection between the local conditional velocity field in Eq. 4 and the pre-image normal-to-tangent transfer operator. Recall from Proposition 2 that the local conditional velocity field under the adopted covariance model is given by $\widehat{v}^c(u_t, t) = v^v(u_t, t) - \frac{1}{1-t}\Sigma_t^v J_t^\top (J_t \Sigma_t^v J_t^\top)^{-1} \mathcal{C}(\mu_t^v)$. Expressing the unconstrained endpoint predictor as $\mu_t^v = u_t + (1-t)v^v(u_t, t)$, this velocity field can be written as $\widehat{v}^c(u_t, t) = (\widehat{\mu}_t^c - u_t)/(1-t)$, where $\widehat{\mu}_t^c \triangleq \mu_t^v - \Sigma_t^v J_t^\top (J_t \Sigma_t^v J_t^\top)^{-1} \mathcal{C}(\mu_t^v)$ denotes the unprojected conditional centroid.

To analyze the correction displacement, consider the linearized normal defect $r_t \triangleq J_t^\top (J_t J_t^\top)^{-1} \mathcal{C}(\mu_t^v) \in N_1$. Multiplying by J_t yields $J_t r_t = J_t J_t^\top (J_t J_t^\top)^{-1} \mathcal{C}(\mu_t^v) = \mathcal{C}(\mu_t^v)$, allowing the displacement to be written as $\Sigma_t^v J_t^\top (J_t \Sigma_t^v J_t^\top)^{-1} \mathcal{C}(\mu_t^v) = \Sigma_t^v J_t^\top (J_t \Sigma_t^v J_t^\top)^{-1} J_t r_t$. Under the flow-induced covariance factorization $\Sigma_t^v = c_t \Phi_t \Phi_t^\top$, the scalar $c_t > 0$ cancels out identically:

$$\Sigma_t^v J_t^\top \left(J_t \Sigma_t^v J_t^\top\right)^{-1} \mathcal{C}(\mu_t^v) = \Phi_t \Phi_t^\top J_t^\top \left(J_t \Phi_t \Phi_t^\top J_t^\top\right)^{-1} J_t r_t. \tag{37}$$

We now project this displacement onto the terminal normal space N_1 and tangent space T_1 using the identity $I_d = P_1 + Q_1$. Recalling the terminal normal projector $Q_1 = J_t^\top (J_t J_t^\top)^{-1} J_t$, its action on the displacement in Eq. 37 satisfies:

$$Q_1 \left[\Phi_t \Phi_t^\top J_t^\top \left(J_t \Phi_t \Phi_t^\top J_t^\top\right)^{-1} J_t r_t\right] = J_t^\top (J_t J_t^\top)^{-1} J_t \Phi_t \Phi_t^\top J_t^\top \left(J_t \Phi_t \Phi_t^\top J_t^\top\right)^{-1} J_t r_t = Q_1 r_t = r_t, \tag{38}$$

where we have used $Q_1 r_t = r_t$ since $r_t \in N_1$. For the tangent component, projecting onto T_1 via P_1 gives $P_1(\Phi_t \Phi_t^\top) Q_1 [Q_1 (\Phi_t \Phi_t^\top) Q_1]^{-1} r_t$. By Proposition 3, the vanishing condition $Q_1 \Phi_t P_t \equiv \mathbf{0}$ ensures that Φ_t is block upper-triangular under the dual basis $(T_t, N_t) \to (T_1, N_1)$. Assuming that the normal propagation operator $C_t \triangleq Q_1 \Phi_t Q_t : N_t \to N_1$ is nonsingular between the m-dimensional subspaces N_t and N_1, the Schur complement in Eq. 8 cancels algebraically:

$$P_1(\Phi_t \Phi_t^\top) Q_1 \left[Q_1 (\Phi_t \Phi_t^\top) Q_1\right]^{-1} = (B_t C_t^\top)(C_t C_t^\top)^{-1} \equiv B_t C_t^{-1}, \tag{39}$$

where $B_t \triangleq P_1 \Phi_t Q_t$. Combining Eq. 38 and Eq. 39 decomposes the displacement into its normal defect and tangent transfer components, yielding the corresponding unprojected endpoint under the adopted covariance model: $\widehat{\mu}_t^c = \mu_t^v - r_t - B_t C_t^{-1} r_t$.

Next, we establish the matrix-free evaluation of the linear inverse $q^* \triangleq C_t^{-1} r_t \in N_t$. Finding q^* is equivalent to solving the linear system $C_t q = r_t$, which can be formulated as the unconstrained quadratic defect energy minimization problem on the pre-image normal bundle:

$$\min_{q \in N_t} f(q) \triangleq \frac{1}{2} \|C_t q - r_t\|^2. \tag{40}$$

The gradient of $f(q)$ on N_t is given by $\nabla_q f(q) = C_t^\top (C_t q - r_t)$. Starting from $q_0 = \mathbf{0} \in N_t$ with initial residual $e_0 = C_t q_0 - r_t = -r_t \in N_1$, at iteration k with iterate q_k and residual $e_k \triangleq C_t q_k - r_t$, the steepest descent search direction is $g_k \triangleq C_t^\top e_k \in N_t$. We perform an exact line search along the ray $q(\gamma) = q_k - \gamma g_k$ to determine the optimal step size:

$$\gamma_k = \arg\min_{\gamma \in \mathbb{R}} \frac{1}{2} \|C_t (q_k - \gamma g_k) - r_t\|^2 = \arg\min_{\gamma \in \mathbb{R}} \frac{1}{2} \|e_k - \gamma C_t g_k\|^2. \tag{41}$$

Differentiating the scalar quadratic objective in Eq. 41 with respect to γ and setting the derivative to zero yields:

$$\frac{d}{d\gamma} \frac{1}{2} \|e_k - \gamma C_t g_k\|^2 = \langle e_k - \gamma C_t g_k,\ -C_t g_k \rangle = 0 \implies \gamma_k \|C_t g_k\|^2 = \langle e_k,\ C_t g_k \rangle. \tag{42}$$

Applying the definition of the adjoint operator, $\langle e_k, C_t g_k \rangle = \langle C_t^\top e_k, g_k \rangle = \langle g_k, g_k \rangle = \|g_k\|^2$. Substituting this into Eq. 42 yields the optimal closed-form step size:

$$\gamma_k = \frac{\|g_k\|^2}{\|C_t g_k\|^2}. \tag{43}$$

With this optimal step size, the energy decreases monotonically, with strict decrease whenever $g_k \neq \mathbf{0}$:

$$f(q_{k+1}) = f(q_k) - \frac{1}{2}\frac{\|g_k\|^4}{\|C_t g_k\|^2} \leq f(q_k). \tag{44}$$

Furthermore, the residual satisfies the linear recurrence $e_{k+1} = C_t q_{k+1} - r_t = C_t(q_k - \gamma_k g_k) - r_t = e_k - \gamma_k C_t g_k$, which updates the residual without requiring an additional operator evaluation.

After K iterations, the iterate $q_{t,K} \in N_t$ provides an approximation to $q^* = C_t^{-1} r_t$, and the corresponding unprojected endpoint is $\mu_t^v - r_t - B_t q_{t,K}$. Applying the terminal Euclidean projection $\Pi_{\mathcal{M}_1}$ maps the corrected endpoint back onto the nonlinear constraint manifold, yielding $z_t^C = \Pi_{\mathcal{M}_1}(\mu_t^v - r_t - B_t q_{t,K})$. Finally, substituting z_t^C into the velocity relation $\widetilde{v}^c(u_t, t) = (z_t^C - u_t)/(1-t)$ establishes Eq. 9, completing the proof. □

A.5 Proof of Proposition 5

Proof. Let $u_t \in \mathbb{R}^d$ be the current state at flow time t, and let $u_0 \sim p_0$ denote the fixed initial sample at $t = 0$. Consider a discrete time step satisfying $0 \leq t < t + \Delta t \leq 1$. The unconstrained one-step forward prediction under the conditional velocity field $\widetilde{v}^c(u_t, t)$ is given by the forward Euler increment:

$$u_{t+\Delta t}^{\text{unconstrained}} \triangleq u_t + \Delta t\, \widetilde{v}^c(u_t, t). \tag{45}$$

Under the variational principle of Minimum Step-Wise Perturbation (MSP), the updated state $u_{t+\Delta t}$ must minimize the Euclidean distance to this unconstrained state $u_{t+\Delta t}^{\text{unconstrained}}$, subject to the requirement that the trajectory remains consistent with a linear displacement path connecting u_0 to an admissible terminal state on the constraint manifold $\mathcal{M}_1 = \{z \in \mathbb{R}^d \mid \mathcal{C}(z) = \mathbf{0}\}$.

Any such admissible state at time level $t + \Delta t$ can be parameterized along the linear displacement path as:

$$u_{t+\Delta t}(z_1) = (1 - t - \Delta t)u_0 + (t + \Delta t)z_1, \qquad z_1 \in \mathcal{M}_1. \tag{46}$$

The variational problem for the discrete update $u_{t+\Delta t}$ is therefore formulated as the constrained minimization:

$$\begin{aligned}\min_{z_1 \in \mathcal{M}_1} \frac{1}{2}\left\|u_{t+\Delta t}(z_1) - u_{t+\Delta t}^{\text{unconstrained}}\right\|^2 = \min_{z_1 \in \mathcal{M}_1} \frac{1}{2}\big\|&(1 - t - \Delta t)u_0 + (t + \Delta t)z_1 \\ &- (u_t + \Delta t\, \widetilde{v}^c(u_t, t))\big\|^2 .\end{aligned} \tag{47}$$

Because $t + \Delta t > 0$, we can factor out the scalar term $(t + \Delta t)$ from the norm:

$$\begin{aligned}&\frac{1}{2}\left\|(t + \Delta t)z_1 - (u_t + \Delta t\, \widetilde{v}^c(u_t, t) - (1 - t - \Delta t)u_0)\right\|^2 \\ &\quad = \frac{1}{2}(t + \Delta t)^2 \left\| z_1 - \frac{u_t + \Delta t\, \widetilde{v}^c(u_t, t) - (1 - t - \Delta t)u_0}{t + \Delta t} \right\|^2 .\end{aligned} \tag{48}$$

Recalling the definition of the equivalent reference endpoint u_1^{ref} from Eq. 12:

$$u_1^{\text{ref}} \triangleq \frac{u_t + \Delta t\, \widetilde{v}^c(u_t, t) - (1 - t - \Delta t)u_0}{t + \Delta t}, \tag{49}$$

the objective function in Eq. 48 simplifies to:

$$\frac{1}{2}(t + \Delta t)^2 \left\|z_1 - u_1^{\text{ref}}\right\|^2 . \tag{50}$$

Since the scalar factor $(t + \Delta t)^2 > 0$ does not alter the location of the minimizer, the variational problem in Eq. 47 is strictly equivalent to minimizing the distance $\|z_1 - u_1^{\text{ref}}\|^2$ over $\mathcal{M}_1$. Assuming

that u_1^{ref} lies in a tubular neighborhood of $\mathcal{M}_1$ where the nearest-point projection is uniquely defined, the minimizer is given by the Euclidean projection:

$$z_1^* = \arg\min_{z_1 \in \mathcal{M}_1} \frac{1}{2} \left\| z_1 - u_1^{\text{ref}} \right\|^2 \triangleq \Pi_{\mathcal{M}_1}\left(u_1^{\text{ref}}\right) = u_1^{\text{corr}}. \tag{51}$$

Substituting the optimal terminal state $z_1^* = u_1^{\text{corr}}$ back into the trajectory parameterization in Eq. 46 directly yields the discrete update rule:

$$u_{t+\Delta t} = (1 - t - \Delta t)u_0 + (t + \Delta t)u_1^{\text{corr}}, \tag{52}$$

establishing Eq. 11.

Finally, we examine the geometric structure of the resulting state perturbation:

$$\begin{aligned} u_{t+\Delta t} - u_{t+\Delta t}^{\text{unconstrained}} &= (1 - t - \Delta t)u_0 + (t + \Delta t)u_1^{\text{corr}} - \left[(1 - t - \Delta t)u_0 + (t + \Delta t)u_1^{\text{ref}}\right] \\ &= (t + \Delta t)\left(u_1^{\text{corr}} - u_1^{\text{ref}}\right). \end{aligned} \tag{53}$$

By the variational characterization of the Euclidean projection onto a C^1 manifold, the displacement vector $u_1^{\text{corr}} - u_1^{\text{ref}} = \Pi_{\mathcal{M}_1}(u_1^{\text{ref}}) - u_1^{\text{ref}}$ is orthogonal to the tangent space of $\mathcal{M}_1$ at u_1^{corr}. Thus, the optimal correction is normal to the terminal constraint manifold at u_1^{corr}, while the tangential component of the reference endpoint is preserved, completing the proof. □

B PRELIMINARIES

In this section, we provide the foundational mathematical framework of Functional Flow Matching (FFM), categorize the representative zero-shot and baseline sampling paradigms, and define the evaluation metrics used throughout our experiments.

B.1 FUNCTIONAL FLOW MATCHING

Functional Flow Matching (FFM) Kerrigan et al. (2023) generalizes continuous normalizing flows to infinite-dimensional Hilbert spaces $\mathcal{U}$ of continuous square-integrable functions $u : \mathcal{X} \to \mathbb{R}^d$ defined over a spatiotemporal domain $\mathcal{X}$. Let μ_0 denote a continuous noise prior measure (e.g., Gaussian Process with a Matérn kernel or white noise) and $\nu = \mu_1$ denote the target data probability measure over the solution space $\mathcal{U}_F$. FFM learns a time-dependent vector field operator $v_\theta : \mathcal{U} \times [1] \to \mathcal{U}$, parameterized by a discretization-invariant Neural Operator (e.g., Fourier Neural Operator), which defines a diffeomorphic flow $\psi_\tau : \mathcal{U} \times [1] \to \mathcal{U}$ via the flow ordinary differential equation (ODE):

$$\frac{\partial}{\partial \tau}\psi_\tau(u) = v_\theta(\psi_\tau(u), \tau), \quad \psi_0(u) = u_0 \sim \mu_0(u), \tag{54}$$

where $\tau \in [1]$ denotes the generative flow time. The flow induces a continuous path of pushforward measures $\hat{\mu}_\tau := (\psi_\tau)_* \mu_0$ connecting μ_0 to $\hat{\mu}_1 \approx \nu$.

Under the Optimal Transport (OT) conditional probability path, samples $u_0 \sim \mu_0$ and $u_1 \sim \mu_1$ are linearly interpolated along straight-line rays:

$$u_\tau = (1 - \tau)u_0 + \tau u_1, \quad \tau \in [1], \tag{55}$$

yielding the tractable conditional Functional Flow Matching objective:

$$\mathcal{L}_{\text{FFM}}(\theta) = \mathbb{E}_{\tau \sim U[1],\, u_0 \sim \mu_0,\, u_1 \sim \mu_1}\left[\left\| v_\theta(u_\tau, \tau) - (u_1 - u_0) \right\|^2_{\mathcal{L}^2(\mathcal{X})}\right]. \tag{56}$$

Once trained, unconstrained sampling proceeds by integrating v_θ from $u_0 \sim \mu_0$ using numerical ODE solvers (e.g., Euler or Dopri5).

B.2 BASELINE MODELS

We compare representative methods for constrained generative sampling, covering zero-shot projection and terminal correction, manifold MCMC, inference-time guidance, and soft-penalty posterior sampling. The following summarizes the mathematical formulation and sampling procedure of each baseline method.

ECI. ECI adopts an extrapolation–correction–interpolation procedure. At time t, the current state u_t is extrapolated to a terminal prediction:

$$\widehat{u}_1 = u_t + (1-t)v_\theta(u_t, t). \tag{57}$$

For an affine constraint $Au = b$, the terminal prediction is corrected using the closed-form Euclidean projection:

$$u_1^{\mathrm{corr}} = \widehat{u}_1 - A^\top (AA^\top)^{-1}(A\widehat{u}_1 - b). \tag{58}$$

For pointwise value constraints, this correction reduces to directly replacing prescribed entries with their target values. The corrected endpoint is then interpolated with the source sample u_0 to obtain the next intermediate state:

$$u_{t'} = (1-t')u_0 + t' u_1^{\mathrm{corr}}. \tag{59}$$

Repeating these three stages propagates constraint information throughout the sampling trajectory.

ECI (nonlinear). ECI (nonlinear) retains the same extrapolation–correction–interpolation structure, replacing the affine correction with a nonlinear manifold projection. Given the terminal prediction $\widehat{u}_1$, the correction is defined as:

$$u_1^{\mathrm{corr}} = \Pi_{\mathcal{M}_1}(\widehat{u}_1), \qquad \mathcal{M}_1 = \{u \mid C(u) = 0\}, \tag{60}$$

where $\Pi_{\mathcal{M}_1}$ denotes the numerical Euclidean nearest-point projection onto the terminal constraint manifold. The subsequent interpolation follows the same update $u_{t'} = (1-t')u_0 + t' u_1^{\mathrm{corr}}$. In our nonlinear geometric benchmarks, this adapted ECI variant utilizes the same geometry-specific projection implementation as PCFM.

PCFM. PCFM combines terminal prediction, tangent-space projection, reverse flow integration, and relaxed constraint correction. Starting from u_t, the flow is integrated forward to obtain $\widehat{u}_1 = \mathrm{ODESolve}(u_t, v_\theta, t, 1)$. Let C denote the constraint residual and $J_1 = \nabla C(\widehat{u}_1)$ its Jacobian. A Gauss–Newton correction gives:

$$u_1^{\mathrm{GN}} = \widehat{u}_1 - J_1^\top (J_1 J_1^\top)^{-1} C(\widehat{u}_1). \tag{61}$$

The corrected endpoint is then propagated backward along the flow-matching displacement path to construct an intermediate reference $\widehat{u}_{t'}$. A relaxed correction incorporates the predicted terminal constraint residual:

$$u_{t'} = \arg\min_w \left\{ \frac{1}{2}\|w - \widehat{u}_{t'}\|_2^2 + \frac{\lambda}{2}\left\|C(w + (1-t')v_\theta(w, t'))\right\|_2^2 \right\}. \tag{62}$$

The procedure alternates these operations throughout sampling, with a final constraint projection when required.

ProFlow. ProFlow alternates between terminal proximal optimization and interpolation along the generative path. First, a terminal prediction is obtained by extrapolating the current state: $\widehat{u}_1 = u_t + (1-t)v_\theta(u_t, t)$. The prediction is refined by minimizing a proximal objective that balances proximity to the flow prediction and physical or observational consistency:

$$u_1^\star = \arg\min_u \left\{ \frac{1}{2}\|u - \widehat{u}_1\|_2^2 + \lambda \mathcal{L}_{\mathrm{cons}}(u) \right\}, \tag{63}$$

where $\mathcal{L}_{\mathrm{cons}}$ denotes the consistency loss used by the proximal formulation. The refined endpoint is subsequently interpolated with the source sample: $u_{t'} = (1-t')u_0 + t' u_1^\star$. This procedure is repeated at successive flow times.

MSP. MSP formulates the discrete update as a minimum-perturbation problem on the linear interpolation path between the source sample and a feasible terminal endpoint. Let $\widetilde{v}_{\mathrm{MSP}}(u_t, t)$ denote the velocity used by the corresponding sampling dynamics. The next state is obtained by:

$$u_{t+\Delta t} = \arg\min_{z_1 \in \mathcal{M}_1} \frac{1}{2}\left\|(1-t-\Delta t)u_0 + (t+\Delta t)z_1 - \left(u_t + \Delta t\, \widetilde{v}_{\mathrm{MSP}}(u_t, t)\right)\right\|_2^2. \tag{64}$$

Equivalently, the corrected endpoint is the projection of the reference endpoint $u_1^{\mathrm{ref}} = \frac{u_t + \Delta t\, \widetilde{v}_{\mathrm{MSP}}(u_t,t) - (1-t-\Delta t)u_0}{t+\Delta t}$, yielding the update:

$$u_{t+\Delta t} = (1-t-\Delta t)u_0 + (t+\Delta t)\Pi_{\mathcal{M}_1}(u_1^{\mathrm{ref}}). \tag{65}$$

CoCoS. CoCoS constructs a Markov chain directly on the constraint manifold $\mathcal{M}_1$ and incorporates the co-area factor into its target measure. For an ambient density p_1, the target distribution is:

$$d\pi^\star(u) \propto p_1(u) \left[\det\big(J(u)J(u)^\top\big)\right]^{-1/2} d\mathcal{H}^{d-m}(u), \qquad u \in \mathcal{M}_1, \tag{66}$$

where $J(u) = \nabla C(u)$ and $\mathcal{H}^{d-m}$ denotes the Hausdorff measure on the manifold. Each transition proposes a displacement in the tangent space, corrects the proposal back onto the manifold, and applies a Metropolis–Hastings accept/reject step using the co-area-corrected target density.

D-Flow. D-Flow guides a pretrained flow by optimizing its initial noise. Let $\Phi_{1\leftarrow 0}$ denote the flow map from the source to the terminal time, and let $\mathcal{L}_C$ denote the terminal constraint loss. Starting from an initial sample u_0, the terminal state is $\Phi_{1\leftarrow 0}(u_0)$, and the source is updated by gradient descent:

$$u_0^{(k+1)} = u_0^{(k)} - \eta \nabla_{u_0} \mathcal{L}_C \left(\Phi_{1\leftarrow 0}(u_0^{(k)})\right). \tag{67}$$

The optimized source is then propagated through the pretrained flow to obtain the constrained generation.

DiffusionPDE-Style Gradient Guidance. Gradient-guided diffusion modifies the reverse sampling dynamics using gradients of a constraint-related loss. Writing the conditional score schematically as:

$$s_{\text{guided}}(u_t, t) = s_\theta(u_t, t) - \lambda_t \nabla_{u_t} \mathcal{L}_C(u_t, t), \tag{68}$$

the guided score is used in the reverse-time diffusion update. Here, λ_t controls the guidance strength, and $\mathcal{L}_C$ denotes the constraint loss.

Unadjusted Langevin Guidance. The unadjusted Langevin sampler targets a soft posterior obtained by combining an ambient prior with a constraint likelihood:

$$\pi_\sigma(u) \propto p_1(u) \exp\left(-\frac{\|C(u)\|_2^2}{2\sigma^2}\right). \tag{69}$$

Its discretized Langevin dynamics are:

$$u^{(k+1)} = u^{(k)} + \eta \nabla_u \log \pi_\sigma(u^{(k)}) + \sqrt{2\eta}\, \xi_k, \qquad \xi_k \sim \mathcal{N}(0, \mathbf{I}). \tag{70}$$

The drift combines the prior score with the gradient of the constraint-induced log-likelihood, while the Gaussian increment provides stochastic exploration.

Soft-Penalty Posterior. The residual-band baseline defines a finite-tolerance target by restricting the ambient prior to samples satisfying a prescribed constraint-residual condition:

$$\pi_\epsilon(u) = Z_\epsilon^{-1} p_1(u) \mathbf{1}(\|C(u)\|_2 < \epsilon), \tag{71}$$

where Z_ϵ is the normalizing constant. Sampling proceeds by drawing candidates from p_1 and accepting those whose constraint residual lies inside the prescribed band.

B.3 Detailed Formulation of Evaluation Metrics

Let $\mathcal{X}$ denote a spatial or space-time grid containing D discretization points. The pointwise empirical means of the generated and reference ensembles are:

$$\bar{u}_{\text{gen},j} = \frac{1}{N_{\text{gen}}} \sum_{n=1}^{N_{\text{gen}}} \hat{u}_j^{(n)}, \qquad \bar{u}_{\text{gt},j} = \frac{1}{N_{\text{gt}}} \sum_{n=1}^{N_{\text{gt}}} u_{\text{gt},j}^{(n)}, \quad j = 1, \ldots, D. \tag{72}$$

The corresponding pointwise empirical standard deviations are denoted by $\sigma_{\text{gen},j}$ and $\sigma_{\text{gt},j}$. Unless otherwise specified, the high-dimensional PDE evaluation uses the sample standard deviation with denominator $N - 1$.

Mean-Field Mean Squared Error (MMSE**).** The MMSE measures the mean squared discrepancy between the generated and reference pointwise mean fields:

$$\mathrm{MMSE} = \frac{1}{D}\sum_{j=1}^{D}\left(\bar{u}_{\mathrm{gen},j} - \bar{u}_{\mathrm{gt},j}\right)^2. \tag{73}$$

For uniformly weighted discrete grids, this is the arithmetic mean of the squared differences over all spatial or space-time grid points. A smaller MMSE indicates more accurate recovery of the posterior mean field.

Standard-Deviation-Field Mean Squared Error (SMSE**).** The SMSE measures the discrepancy between the pointwise standard deviation fields:

$$\mathrm{SMSE} = \frac{1}{D}\sum_{j=1}^{D}\left(\sigma_{\mathrm{gen},j} - \sigma_{\mathrm{gt},j}\right)^2. \tag{74}$$

A smaller SMSE indicates more accurate recovery of the spatial uncertainty structure. For the high-dimensional PDE experiments, both MMSE and SMSE are computed as the arithmetic means of the corresponding squared pointwise differences over the flattened solution fields.

Mean Vector Error (mean err). For the finite-dimensional Darcy Flow inverse problem, let $\hat{\mu}_{\mathrm{gen}} \in \mathbb{R}^d$ and $\hat{\mu}_{\mathrm{gt}} \in \mathbb{R}^d$ denote the empirical mean vectors of the generated and reference ensembles, respectively. The mean vector error is the Euclidean distance between the generated and reference posterior means:

$$\mathrm{Err}_{\mathrm{mean}} = \left\|\hat{\mu}_{\mathrm{gen}} - \hat{\mu}_{\mathrm{gt}}\right\|_2. \tag{75}$$

Relative Standard-Deviation Error (std rel. err). Let $\hat{\sigma}_{\mathrm{gen}} \in \mathbb{R}^d$ and $\hat{\sigma}_{\mathrm{gt}} \in \mathbb{R}^d$ denote the empirical standard deviation vectors. The relative standard-deviation error is computed as the arithmetic mean of the coordinate-wise relative absolute errors:

$$\mathrm{RelErr}_{\mathrm{std}} = \frac{1}{d}\sum_{j=1}^{d}\frac{|\hat{\sigma}_{\mathrm{gen},j} - \hat{\sigma}_{\mathrm{gt},j}|}{\hat{\sigma}_{\mathrm{gt},j}}. \tag{76}$$

This definition matches the numerical implementation, which averages the coordinate-wise absolute differences normalized by the corresponding reference standard deviations.

90% Marginal Credible-Interval Coverage (Cov_{90}**).** For each coordinate j, we construct a central 90% interval from the empirical 5th and 95th percentiles of the generated ensemble:

$$I_j^{90} = \left[q_{0.05,j}^{\mathrm{gen}},\, q_{0.95,j}^{\mathrm{gen}}\right], \tag{77}$$

where $q_{\alpha,j}^{\mathrm{gen}}$ is the empirical α-quantile of the generated samples in coordinate j. Using an independent reference ensemble, the empirical marginal coverage is:

$$\mathrm{Cov}_{90} = \frac{1}{dN_{\mathrm{gt}}}\sum_{j=1}^{d}\sum_{n=1}^{N_{\mathrm{gt}}}\mathbb{I}\left[u_{\mathrm{gt},j}^{(n)} \in I_j^{90}\right]. \tag{78}$$

The nominal value is 0.90. This metric measures average coordinate-wise marginal coverage, rather than joint coverage of a high-dimensional credible region.

One-Dimensional Wasserstein-1 Distance (W_1**).** For two probability distributions P and Q on the real line, the Wasserstein-1 distance is:

$$W_1(P,Q) = \int_{-\infty}^{+\infty}\left|F_P(z) - F_Q(z)\right|\,dz, \tag{79}$$

where F_P and F_Q are their cumulative distribution functions. In the two-dimensional synthetic manifold benchmarks, this distance is evaluated along the scalar parameterization of the constrained manifold.

Average Marginal Wasserstein-1 Distance ($\overline{W}_1$). In the Darcy Flow benchmark, the reported metric is the average of the one-dimensional Wasserstein-1 distances between the corresponding marginal distributions. Writing $P_{\mathrm{gen}}^{(j)}$ and $P_{\mathrm{gt}}^{(j)}$ for the marginal distributions of the generated and reference distributions, respectively, the metric is defined as:

$$\overline{W}_1\left(P_{\mathrm{gen}}, P_{\mathrm{gt}}\right) = \frac{1}{d}\sum_{j=1}^{d} W_1\left(P_{\mathrm{gen}}^{(j)}, P_{\mathrm{gt}}^{(j)}\right). \tag{80}$$

This quantity measures the average marginal discrepancy and is not the joint multivariate Wasserstein-1 distance. The empirical cumulative distribution functions are approximated by linearly interpolating the sorted samples on a uniform grid of 500 points spanning the combined sample range.

Kullback–Leibler Divergence (D_{KL}). The Kullback–Leibler (KL) divergence is used in the two-dimensional synthetic posterior benchmark to evaluate the mismatch between the reference and generated densities along the scalar manifold parameterization. The density estimates are discretized into B bins. Let $P_{\mathrm{gt}}(b)$ and $P_{\mathrm{gen}}(b)$ denote the normalized probability masses in bin b:

$$D_{\mathrm{KL}}\left(P_{\mathrm{gt}} \parallel P_{\mathrm{gen}}\right) = \sum_{b=1}^{B} P_{\mathrm{gt}}(b)\log\left(\frac{P_{\mathrm{gt}}(b)}{P_{\mathrm{gen}}(b)+\epsilon}\right), \qquad \epsilon = 10^{-12}. \tag{81}$$

The stabilizing constant prevents division by zero when a generated histogram bin has zero probability mass. Smaller values indicate closer agreement with the reference density.

Fréchet Poseidon Distance (FPD). The Fréchet Poseidon Distance (FPD) evaluates distributional alignment in the feature space of the pretrained Poseidon PDE foundation model. For each solution field, the input is converted to a single-channel tensor when necessary and resized to 128×128 by bilinear interpolation. When the Poseidon-B (157M) model is successfully loaded, we extract its encoder output using a forward hook, flatten the output, and retain the first 784 feature components. Let z_{gen} and z_{gt} denote the resulting 784-dimensional embeddings for generated and reference samples. Their empirical means and covariance matrices are denoted by $(\mu_{\mathrm{gen}}^z, \Sigma_{\mathrm{gen}}^z)$ and $(\mu_{\mathrm{gt}}^z, \Sigma_{\mathrm{gt}}^z)$, respectively. The FPD is computed as:

$$\mathrm{FPD} = \left[\left\|\mu_{\mathrm{gen}}^z - \mu_{\mathrm{gt}}^z\right\|_2^2 + \mathrm{Tr}(\Sigma_{\mathrm{gen}}^z) + \mathrm{Tr}(\Sigma_{\mathrm{gt}}^z) - 2\,\mathrm{Tr}\left(\left(\Sigma_{\mathrm{gen}}^z\Sigma_{\mathrm{gt}}^z\right)^{1/2}\right)\right]_+^{1/2}, \tag{82}$$

where $[a]_+ = \max(a, 0)$ prevents a negative distance squared caused by numerical roundoff. If the matrix square-root routine returns non-finite values, the implementation retries after adding $\epsilon\mathbf{I}$ to both covariance matrices with $\epsilon = 10^{-6}$. Lower values indicate closer alignment between the generated and reference feature distributions.

Wall-Clock Sampling Time (Time). The sampling time is measured using wall-clock time around the sampling call for each method and is reported in seconds. It excludes construction of the Gold reference ensemble and the subsequent distribution-metric computations. Runtime comparisons are interpreted under the recorded hardware, sample-count, and numerical settings.

C More Details of Uncertainty Quality Benchmark

In this section, we provide full technical details for the Uncertainty Quantification (UQ) benchmark setup, including the governing equations, observation operators, the gold-standard reference posterior generation protocol, baseline hyperparameter configurations, and statistical sampling stability.

C.1 Governing PDE Systems and Parameterization

We evaluate our framework on a representative steady-state nonlinear Darcy flow inverse problem defined over the spatial domain $\mathcal{D} = [0, 1]$. The governing equation is given by the one-dimensional elliptic partial differential equation (PDE):

$$-\frac{d}{dx}\left(\kappa(x;\theta)\frac{dh}{dx}\right) = 0, \quad x \in (0,1), \tag{83}$$

subject to Dirichlet boundary conditions $h(0) = 1.0$ and $h(1) = 0.0$, where $h(x)$ represents the hydraulic head and $\kappa(x;\theta)$ denotes the spatially varying permeability field.

The log-permeability field $\log \kappa(x;\theta)$ is parameterized via an 8-dimensional Karhunen-Loève (KL) style truncated expansion over spatial coordinates:

$$\log \kappa(x;\theta) = \sum_{j=1}^{d} \theta_j \phi_j(x) S_j, \quad d = 8, \tag{84}$$

where $\phi_j(x) = \cos(j\pi x)$ denotes the spatial basis functions, $S_j = 1.2/j$ represents the spectral decay coefficients, and the latent parameter vector $\theta \in \mathbb{R}^d$ follows an isotropic standard Gaussian prior distribution, i.e., $p(\theta) = \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.

C.2 Observation Protocol and Constraint Formulation

To simulate sparse internal pressure (hydraulic head) monitoring, we establish an observational operator $\mathcal{G} : \mathbb{R}^d \to \mathbb{R}^m$ that solves the Darcy equation and extracts head values at $m = 3$ localized spatial sensors situated at grid indices corresponding to $x \in \{12, 24, 36\}$ out of $N_{\text{grid}} = 49$ uniform collocation nodes.

Let $\mathcal{H} : \mathbb{R}^m \to \mathbb{R}^m$ represent the measurement mapping. Given a set of synthetic ground-truth observations $y_{\text{obs}} = \mathcal{H}(\mathcal{G}(\theta_{\text{true}}))$ generated from a reference parameter vector θ_{true}, the physical consistency and observation consistency define the exact zero-residual terminal hard constraint operator:

$$C(\theta) = \mathcal{H}(\mathcal{G}(\theta)) - y_{\text{obs}} = \mathbf{0}, \quad C : \mathbb{R}^d \to \mathbb{R}^m, \tag{85}$$

which restricts the ambient prior distribution onto a measure-zero feasible manifold $\mathcal{M}_1 = \{\theta \in \mathbb{R}^d \mid C(\theta) = \mathbf{0}\}$.

C.3 Gold-Standard Reference Posterior Construction

Because obtaining the true Bayesian conditional posterior $\pi^\star(\theta)$ in nonlinear PDE inverse problems is analytically intractable, we construct an unbiased ground-truth (GT) reference arbiter via high-throughput rejection sampling from the isotropic Gaussian prior $p(\theta)$.

Specifically, we draw independent latent parameter vectors $\theta \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ and accept samples that strictly satisfy the tight residual threshold $\|C(\theta)\|_2 < \varepsilon$, where the tolerance is set to $\varepsilon = 0.006$. We collect a total of $N = 40{,}000$ independent accepted samples. To rigorously estimate the empirical sampling-noise floor (W_1^{floor}), this gold-standard pool is randomly partitioned into two equal, disjoint subsets (GT_1 and GT_2, with $20{,}000$ samples each) to compute the baseline Wasserstein-1 discrepancy limit.

C.4 Baseline Implementation and Sampling Hyperparameters

We compare four distinct inference paradigms under identical evaluation batches of size $B = 6{,}000$:

- **Manifold MCMC (CoCoS):** Evaluated via the ZHG manifold MCMC sampler operating directly on $\mathcal{M}_1$. The chains run for 600 iterations with 300 burn-in steps, a thinning interval of 10, and a Gaussian proposal step size $s = 0.10$. **CoCoS** incorporates the analytical co-area Fixman determinant correction $\log\det\left(JJ^\top\right)^{-1/2}$.
- **Inference-time Guidance (Guided Langevin):** Represents the gradient-guidance mechanism shared by D-Flow and DiffusionPDE. It employs an unadjusted Langevin diffusion sampler targeting the soft posterior with noise variance $\sigma = 0.02$, running for $4{,}000$ iterations with a step size of $\eta = 2 \times 10^{-4}$.
- **Soft-penalty Posterior:** Represents the finite-noise training target (analogous to PBFM/PIDM training objectives). It performs direct rejection sampling from the Gaussian prior under a wider residual band of $\|C(\theta)\|_2 < 0.02$.
- **Finite-Step Projection and Flow Methods (PCFM, CMSP):** Both deterministic trajectory methods are evaluated with a uniform time discretization of $N_{\text{steps}} = 30$ integration steps from $t = 0$ to

$t = 1$. **PCFM** employs standard Euclidean Gauss-Newton projections. **CMSP (Ours)** additionally incorporates Pre-Image Invariant Spectral Transfer (PIS) with $K = 3$ exact line-search iterations per flow step.

C.5 Sample Size and Statistical Stability

To ensure that empirical UQ metrics (such as MMSE, $\text{RelErr}_{\text{std}}$, and Cov_{90}) are free from finite-sample estimation variance, all generative and MCMC evaluators extract a standardized evaluation batch of size $N_{\text{eval}} = 6{,}000$ independent valid samples after constraint cleaning. All numerical experiments are executed under a fixed global random seed (SEED $= 0$).

D More Details and Experiments of Posterior Sweep

D.1 Basic Setup

To ensure a fair and rigorous comparison across all geometric configurations, all synthetic 2D manifold experiments share identical ambient prior-to-target probability paths, generative dynamics, and evaluation protocols.

Ambient Generative Setup and Prior Flow. The ambient space is fixed to $\mathbb{R}^2$, where the generative flow transports an isotropic standard Gaussian source $p_0 = \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$ to an anisotropic, correlated target Gaussian $p_1 = \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ with mean $\boldsymbol{\mu}_1 = [0.2, 2.5]^\top$ and covariance $\boldsymbol{\Sigma}_1 = [[0.45, 0.28], [0.28, 0.35]]$. Crucially, we evaluate the unconstrained velocity field $v^v(u, t)$ and its spatial flow Jacobian analytically via optimal transport displacement interpolation, eliminating neural network parameterization errors and ensuring that all empirical performance discrepancies stem strictly from the respective conditioning and projection mechanisms.

Sample Scales and Discretization Trajectories. Initial particles $u_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$ are sampled independently using a fixed random seed ($\texttt{SEED} = 42$). For the representative parabolic benchmark discussed in the main text, we simulate $N_{\text{samples}} = 2{,}000$ particles over $N_{\text{steps}} = 40$ uniform Euler integration steps. For the remaining extended geometric manifolds (ellipse, compound wave, and singular curve), we evaluate $N_{\text{samples}} = 600$ independent trajectories with $N_{\text{steps}} = 30$ steps across all compared paradigms.

Probability Mass Percentages and Adaptive Center Calibration. To systematically test whether samplers can adapt to highly asymmetric posterior distributions, each manifold family is evaluated across five standardized mass-ratio configurations: $\mathcal{Q} = \{\text{Q95}, \text{Q75}, \text{Q50}, \text{Q25}, \text{Q5}\}$, where Q$p$ specifies that exactly $p\%$ of the true conditional probability mass lies on the designated left sub-domain of the manifold (e.g., $x \leq c$ or $\theta \in [\pi/2, 3\pi/2]$). Because the ambient prior is tilted and asymmetric, the geometric translation center $c \in \mathbb{R}$ is dynamically calibrated for each target percentage via Brent's root-finding method with tolerance $\varepsilon \leq 10^{-10}$. The calibrated center coordinates and realized empirical mass ratios for all benchmarks are cataloged in Table 5.

Table 5: **Calibrated horizontal translation centers (c) across four geometric manifolds.** Evaluated at five target left-domain posterior probability mass percentages ($\mathcal{Q} = \{\text{Q95}, \text{Q75}, \text{Q50}, \text{Q25}, \text{Q5}\}$). Each center c is solved dynamically via root-finding to strictly match the specified prior truncation.

Manifold Geometry	**Q95** (95%)	**Q75** (75%)	**Q50** (50%)	**Q25** (25%)	**Q5** (5%)
Translated Parabola	-0.425	-0.779	-0.992	-1.189	-1.481
Closed Ellipse	$+1.311$	$+0.714$	$+0.200$	-0.314	-0.911
Multi-Frequency Wave	$+0.656$	$+0.384$	$+0.195$	-0.029	-0.785
Rational Singular Curve	$+0.480$	-0.059	-0.389	-0.729	-1.349

D.2 Reason of Tasks Selection

Translated Parabola. We first evaluate our framework on the translated convex parabola, defined by the equality constraint:

$$C(u) = u_2 - \left[\kappa(u_1 - c)^2 + \delta\right] = 0, \tag{86}$$

where the curvature coefficient is set to $\kappa = 0.2$, the vertical offset is $\delta = 0.8$, and $c \in \mathbb{R}$ denotes the dynamically calibrated translation center.

Geometrically, this manifold is unbounded and strictly convex, with tangent slopes that grow linearly with the distance from the apex while its positive curvature decays monotonically toward the tails. This setup induces severe measure-theoretic challenges primarily through two mechanisms: first, a pronounced Gaussian–manifold misalignment, in which the tilted, correlated contours of the ambient prior intersect nonlinearly with the parabolic arc to create dynamic posterior peaks, shifting mass partitions, and asymmetric tails as the vertex translates; and second, a spatially varying co-area factor governed by $[1 + 4\kappa^2(u_1 - c)^2]^{-1/2}$, which demands rigorous compensation for the quadratic decay of the geometric volume element away from the apex.

Because uncorrected baseline methods apply Euclidean nearest-point projections relative to an isotropic metric, they systematically overlook the prior covariance structure and the non-uniform volume distortion, resulting in substantial mode displacements, spurious probability over-concentration near the vertex, and severe tail truncation across varying mass-ratio regimes.

Physically, this unidirectional convex geometry serves as an idealized benchmark for asymmetric potential wells, curved impermeable boundary layers in subterranean Darcy flows, and directional thermal transport fronts.

Closed Ellipse. We next evaluate our framework on the horizontally translated closed ellipse, defined by the quadratic constraint:

$$C(u) = \left(\frac{u_1 - c}{a}\right)^2 + \left(\frac{u_2 - y_0}{b}\right)^2 - 1 = 0, \tag{87}$$

where the horizontal semi-axis is $a = 1.8$, the vertical semi-axis is $b = 1.1$, the fixed vertical center is $y_0 = 2.5$, and $c \in \mathbb{R}$ represents the calibrated horizontal translation center.

Geometrically, this manifold constitutes a bounded, smooth, and compact one-dimensional closed curve with periodic topology, lacking boundaries or asymptotic endpoints. This geometry presents distinct measure-theoretic challenges spanning three aspects: first, the requirement of global mass allocation under a compact topology, where generated samples must correctly distribute along the entire closed perimeter to accurately capture both high-density modes and low-density tails without boundary escape; second, an exact co-area cancellation, where under the angular parameterization $u_1(\theta) = c + a\cos\theta$ and $u_2(\theta) = y_0 + b\sin\theta$, the product of the differential arc-length and the co-area volume factor $[\|\nabla C\|]^{-1}$ evaluates to the constant $ab/2$, thereby eliminating spatially varying geometric measure weights and isolating the model's intrinsic ability to condition on the anisotropic density itself; and third, the strong coupling between prior anisotropy and rotating tangent-normal frames, where the tilted Gaussian contours interact dynamically with the constantly rotating curvature, producing a highly non-uniform annular probability density.

Because uncorrected baseline methods apply Euclidean projections driven solely by geometric distance, they fail to preserve the target mass across different quadrants when the prior covariance misaligns with the local manifold curvature, yielding systematic inter-regional mass leakage, mode displacements, and distorted relative probability masses.

Physically, this compact periodic geometry benchmarks scientific systems constrained by closed boundaries, such as circulating fluid flows in closed conduits, magnetic flux surfaces in toroidal confinement devices, and conservative orbital mechanics.

Multi-Frequency Compound Wave. We further evaluate our framework on the translated multi-frequency oscillatory wave, defined by the explicit function constraint:

$$C(u) = u_2 - [y_{\text{base}} + s(u_1 - c) + A_1\sin(\omega_1(u_1 - c)) + A_2\sin(\omega_2(u_1 - c))] = 0, \tag{88}$$

where the baseline height is $y_{\text{base}} = 1.2$, the global linear drift is $s = -0.22$, the low-frequency component has amplitude $A_1 = 0.82$ and frequency $\omega_1 = 1.25$, the high-frequency harmonic has amplitude $A_2 = 0.23$ and frequency $\omega_2 = 3.6$, and $c \in \mathbb{R}$ represents the horizontal shift center.

Geometrically, this manifold forms an open, multiscale oscillatory curve characterized by alternating positive and negative curvatures with densely distributed local extrema and inflection points. This geometry introduces intricate measure and dynamics challenges across two primary dimensions: first, the formation of multiscale local modes, where the superposition of low-frequency carrier waves and high-frequency ripples creates densely spaced local valleys and crests, substantially increasing the difficulty of properly resolving localized posterior probability mass; and second, the rapid alternation of curvature and normal directions, where adjacent crests and troughs continuously flip the direction of the normal space $\mathcal{N}_t$, demanding that the conditioning vector field dynamically adapt to alternating local geometries rather than relying on a unidirectional geometric correction.

Because uncorrected baseline methods apply isotropic Euclidean projections, discrete integration steps readily overshoot sharp local peaks and falsely capture particles into adjacent non-target troughs, causing severe mode misalignments, complete omission of localized modes, and spurious multi-modal artifacts that fail to recover the true underlying density.

Physically, this oscillatory multiscale geometry directly mirrors complex physical phenomena such as high-frequency acoustic wave propagation, corrugated fluid interfaces induced by Kelvin–Helmholtz instabilities, rough geological fracture zones, and multiscale boundary interfaces in porous media.

Rational Singular Curve. Finally, we evaluate our framework on the horizontally translated rational singular curve, defined by the level set:

$$C(u) = u_2 - \frac{(u_1 - c)^2}{[1 - (u_1 - c)]\,[1 + (u_1 - c)]^3} = 0, \tag{89}$$

where $c \in \mathbb{R}$ is the calibrated translation center, and safety neighborhoods $|u_1 - c \pm 1| \leq \epsilon_{\text{sing}}$ with $\epsilon_{\text{sing}} = 10^{-2}$ are excluded to avoid unbounded evaluation across the vertical poles.

Geometrically, this manifold constitutes a non-compact, multi-branch geometry split into three disconnected components by vertical asymptotic singularities at $u_1 - c = -1$ (a third-order pole) and $u_1 - c = 1$ (a first-order pole). This topology imposes extreme theoretical and algorithmic demands across two critical fronts: first, numerical and gradient instabilities near singular points, where the constraint gradient $\|\nabla C\|$ explodes dramatically as trajectories approach the poles, creating localized zones of extreme geometric sensitivity that severely challenge discrete integration steps and linear transfers; and second, global cross-branch probability mass allocation, where the generative model must not only reconstruct the continuous conditional distribution within each isolated branch, but also accurately partition the discrete probability measure across the disconnected components according to the ambient Bayesian prior.

Because uncorrected baseline methods perform projections governed purely by isotropic Euclidean distances, particles are aggressively pulled toward whichever branch is geometrically closest in Euclidean space rather than probabilistically consistent under the prior, precipitating severe cross-branch mass imbalance, total probability collapse on distant branches, and catastrophic numerical divergence near the asymptotic boundaries.

Physically, this singular multi-branch geometry provides an ideal mathematical testbed for modeling physical systems governed by steep gradients and topological discontinuities, such as shock waves and contact discontinuities in compressible aerodynamics, the sonic barrier in transonic flows, and stress singularity zones near crack tips in fracture mechanics.

D.3 Numerical Realization of Manifold Projections

Across all evaluated generative paradigms and geometric benchmarks, the manifold projection operator is mathematically defined as the Euclidean nearest-point projection onto the terminal constraint level set:

$$\Pi_{\mathcal{M}_1}(z) = \arg\min_{u \in \mathcal{M}_1} \frac{1}{2}\|u - z\|_2^2. \tag{90}$$

While all compared methods share identical projection operators within each benchmark to guarantee absolute fairness, the numerical solver is tailored to the specific topological and analytical properties of each manifold family.

Translated Parabola (Analytical Stationarity Solver). For a query point $z = (x_0, y_0)$, parameterizing the parabola via the relative coordinate $r = x - c$ yields the squared distance objective $D(r) = (r - (x_0 - c))^2 + (\kappa r^2 + \delta - y_0)^2$. Setting the first derivative with respect to r to zero yields the exact cubic stationarity equation:$2\kappa^2 r^3 + [1 + 2\kappa(\delta - y_0)]\, r - (x_0 - c) = 0$. We solve for all roots of this cubic polynomial analytically via companion matrix eigenvalues, filter the valid real roots, and select the stationary candidate that globally minimizes $D(r)$. A constrained Sequential Least Squares Programming (SLSQP) routine serves as a machine-precision numerical fallback.

Multi-Frequency Wave (Grid Search with Damped Newton Refinement). Parameterizing the oscillatory curve via its Cartesian graph $(x, f(x))$, we first execute a dense grid search over $x \in [X_{\min}, X_{\max}]$ using $1{,}201$ nodes to identify the optimal initial basin of attraction. Starting from this global grid candidate, we perform up to 10 iterations of a second-order Newton-Raphson scheme equipped with an 8-step backtracking line search (restricting step updates to $|\Delta x| \leq 0.25$) to converge strictly onto the zero-gradient stationary point.

Closed Ellipse (Periodic Angular Newton Solver). To naturally accommodate the multi-valued branches of the closed boundary, the ellipse is parameterized by its continuous angular coordinate $q(\theta) = (c + a\cos\theta, y_0 + b\sin\theta)$ for $\theta \in [0, 2\pi)$. We initialize θ by evaluating distances across an angular grid of $1{,}201$ uniform bins, followed by up to 10 iterations of angular Newton refinement with modulo-2π periodic phase wrapping and backtracking line search (bounded by $|\Delta\theta| \leq \pi/4$).

Rational Singular Curve (Branch-Restricted Pole-Excluded Solver). Due to the vertical asymptotic poles at $s = x - c = \pm 1$, the feasible domain is partitioned into three disjoint connected components outside the safety radius $\epsilon_{\text{sing}} = 10^{-2}$. We first deploy dense grid evaluations across all three branches (801 nodes per segment) to determine the globally nearest branch and initial candidate. Subsequently, we execute up to 10 iterations of damped Newton-Raphson refinement strictly clamped within the bounds of the chosen branch, preventing unphysical trajectory jumping across the singular poles.

Crucially, within any given task, all models including ECI (nonlinear) and our proposed CMSP invoke the exact same underlying projection operator, ensuring that all comparative differences stem entirely from the sampling dynamics and measure-conditioning mechanisms.

D.4 Results Analysis

Table 6 summarizes the results across four geometric manifolds and five posterior mass ratios. CMSP achieves the best performance in 51 out of 60 evaluations (85.0%), consistently outperforming competing methods in recovering posterior moments and overall distributional structure. Its advantage persists across diverse geometric configurations and varying posterior mass allocations, demonstrating robust performance in both moment-level accuracy and distributional fidelity. These results highlight the effectiveness of integrating geometric measure correction with conditional transport to approximate the target posterior across challenging constrained sampling tasks.

For detailed qualitative and quantitative visual inspections on the primary translated parabola task, the quantitative metric bar plots, fine-grained posterior probability density profiles, and multi-task violin distributions of relative displacements are depicted in Fig. 5, Fig. 1, and Fig. 12, respectively. Correspondingly, the complete suites of evaluation figures for the remaining three manifold geometries are systematically presented in Figs. 6–15.

Table 6: Comprehensive benchmark across four geometric manifolds and five posterior mass ratios. **Bold gold** indicates the best performance, and **standard bold** indicates the second.

Method	Mean Field MSE (MMSE) ↓					Standard Deviation Field MSE (SMSE) ↓					Wasserstein-1 Distance (W_1) ↓				
	Q95	Q75	Q50	Q25	Q5	Q95	Q75	Q50	Q25	Q5	Q95	Q75	Q50	Q25	Q5
Translated Parabola															
PCFM	2.498	2.986	2.264	1.022	0.083	11.581	5.364	1.889	0.569	0.056	1.546	1.630	1.369	0.925	0.305
ECI (nonlinear)	2.191	**2.772**	**2.144**	**0.976**	**0.079**	**10.089**	**4.941**	**1.771**	**0.537**	**0.052**	**1.443**	**1.564**	**1.325**	**0.898**	**0.287**
ProFlow	**2.113**	3.410	2.765	1.340	0.165	10.290	6.562	2.480	0.816	0.129	1.457	1.820	1.650	1.281	0.779
MSP	3.358	3.495	2.535	1.123	0.090	15.751	6.370	2.150	0.639	0.064	1.802	1.777	1.463	0.982	0.345
CMSP (**Ours**)	**0.389**	**0.672**	**0.515**	**0.138**	**0.015**	**1.877**	**1.249**	**0.439**	**0.078**	**0.009**	**0.624**	**0.785**	**0.650**	**0.349**	**0.096**
Closed Ellipse															
PCFM	0.073	0.113	5.7E-04	0.098	0.080	0.135	0.065	**1.2E-07**	0.070	0.167	0.211	0.246	0.168	0.224	0.235
ECI (nonlinear)	0.077	0.120	**5.5E-04**	0.103	0.084	0.151	0.074	**3.8E-06**	0.077	0.183	0.223	0.254	**0.154**	0.233	0.246
ProFlow	**0.048**	**0.030**	8.9E-04	**0.020**	**0.050**	0.145	**2.0E-04**	3.2E-04	**0.001**	0.143	0.221	0.361	0.421	0.333	0.223
MSP	0.065	0.097	6.7E-04	0.085	0.072	**0.104**	0.047	6.1E-05	0.057	**0.136**	**0.186**	**0.242**	0.214	**0.218**	**0.212**
CMSP (**Ours**)	**0.029**	**0.042**	**4.6E-05**	**0.040**	**0.039**	**0.083**	**0.038**	4.1E-06	**0.039**	**0.113**	**0.164**	**0.172**	**0.034**	**0.161**	**0.193**
Multi-Frequency Compound Wave															
PCFM	2.664	1.635	0.641	0.120	0.027	6.577	1.614	0.545	0.147	0.220	1.560	1.190	0.721	0.322	0.163
ECI (nonlinear)	**2.653**	**1.629**	**0.639**	**0.118**	**0.023**	**6.542**	**1.604**	**0.542**	**0.142**	0.177	**1.556**	**1.187**	**0.720**	**0.320**	**0.148**
ProFlow	3.974	2.458	1.061	0.253	0.038	9.954	2.432	0.891	0.300	**0.163**	1.920	1.477	0.955	0.495	0.214
MSP	2.673	1.641	0.643	0.122	0.042	6.606	1.626	0.551	0.153	0.352	1.564	1.194	0.723	0.324	0.206
CMSP (**Ours**)	**1.105**	**0.818**	**0.297**	**0.041**	**0.008**	**2.733**	**0.786**	**0.235**	**0.036**	**0.032**	**1.009**	**0.836**	**0.482**	**0.197**	**0.116**
Rational Singular Curve															
PCFM	0.120	0.316	0.223	0.111	0.183	**0.648**	0.434	0.193	**0.051**	**0.010**	0.283	0.428	0.330	0.143	**0.033**
ECI (nonlinear)	0.122	0.317	**0.222**	**0.108**	**0.165**	**0.648**	0.434	0.193	**0.051**	**0.010**	0.283	0.428	0.330	0.143	**0.032**
ProFlow	**0.024**	0.469	0.552	0.262	0.220	**1.3E-04**	0.743	0.664	0.315	0.079	**0.017**	0.561	0.607	0.355	0.087
MSP	0.117	**0.314**	0.224	0.121	0.237	0.649	**0.433**	**0.192**	**0.051**	0.011	**0.283**	**0.427**	**0.329**	**0.143**	0.034
CMSP (**Ours**)	**0.040**	**0.053**	**0.028**	**0.020**	**0.001**	**0.012**	**0.042**	**0.004**	**0.002**	**3.4E-04**	**0.137**	**0.142**	**0.053**	**0.089**	0.058

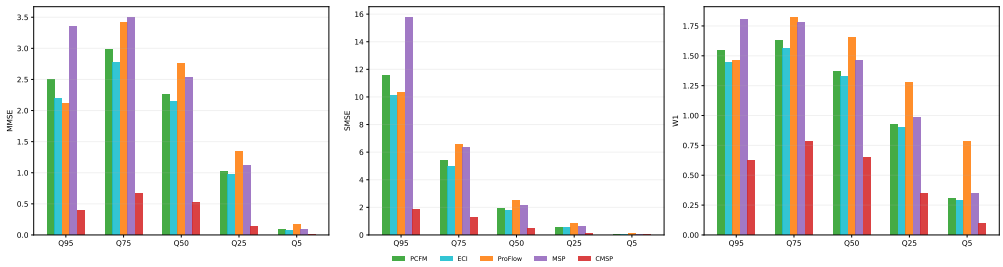


Figure 5: Quantitative metrics on the translated parabola task.

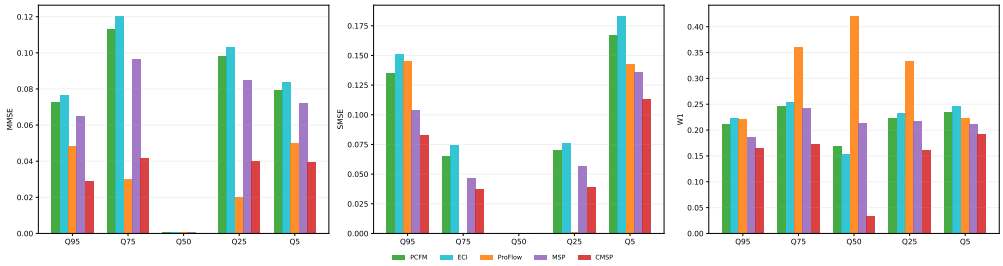


Figure 6: Quantitative metrics on the closed ellipse task.

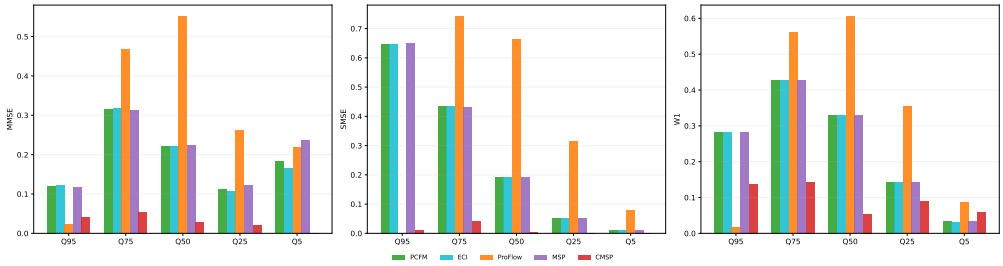


Figure 8: Quantitative metrics on the rational singular curve task.

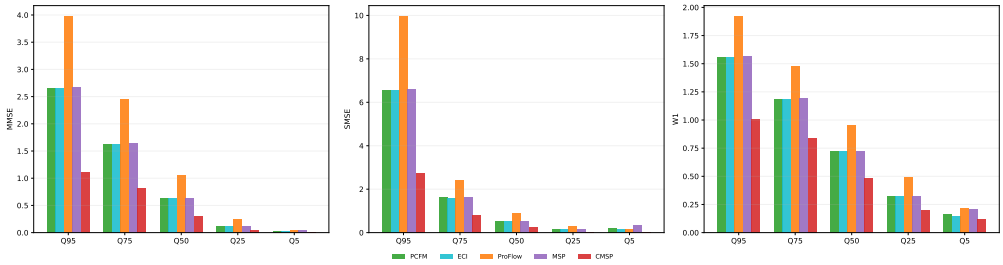


Figure 7: Quantitative metrics on the multi-frequency wave task.

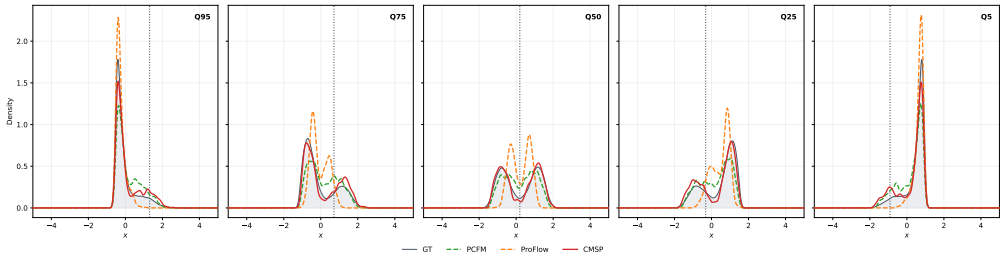


Figure 9: Posterior probability density profiles on the closed ellipse task. Shaded gray area represents the ground truth (GT); vertical dotted line indicates the translation center c.

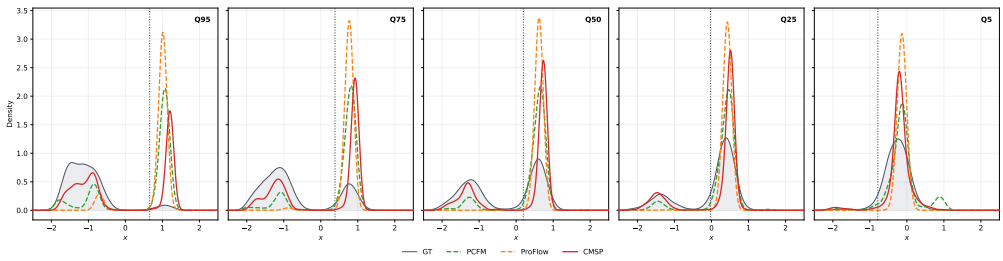


Figure 10: Posterior probability density profiles on the multi-frequency wave task. Shaded gray area represents the ground truth (GT); vertical dotted line indicates the translation center c.

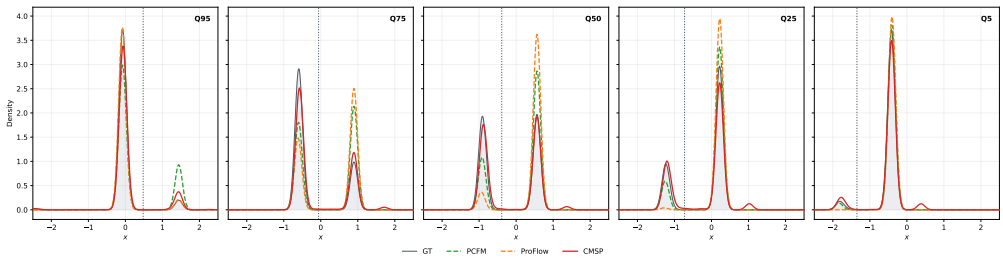


Figure 11: Posterior probability density profiles on the rational singular curve task. Shaded gray area represents the ground truth (GT); vertical dotted line indicates the translation center c.

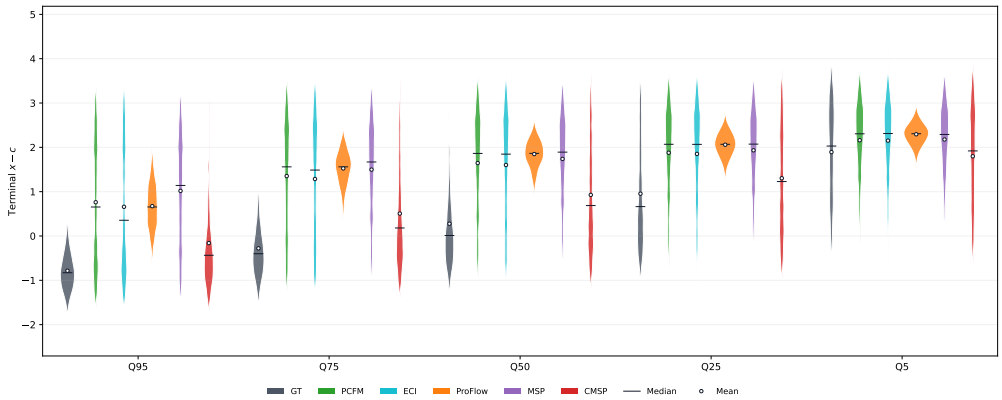


Figure 12: Violin distributions of terminal relative displacements ($x - c$) on the parabola task. Black lines and white dots denote the median and mean, respectively.

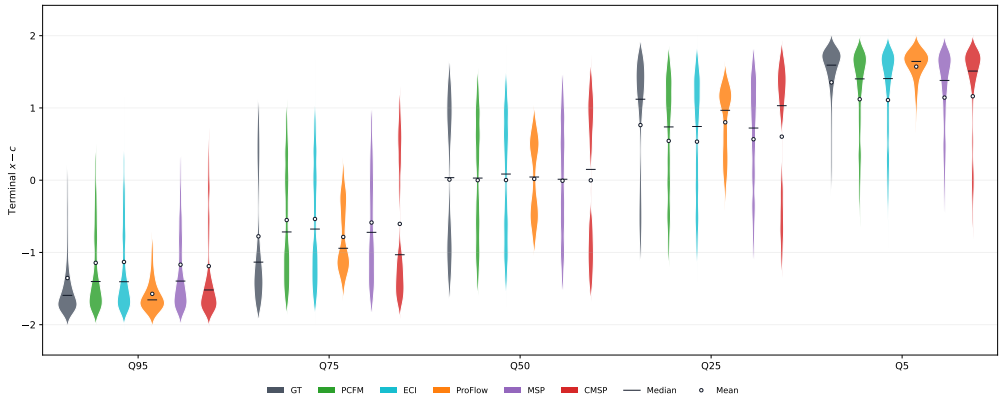


Figure 13: Violin distributions of terminal relative displacements ($x - c$) on the closed ellipse task. Black lines and white dots denote the median and mean, respectively.

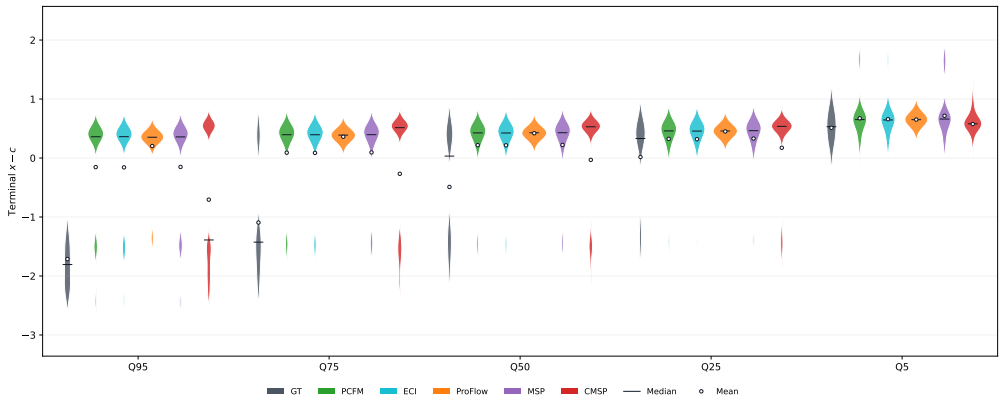


Figure 14: Violin distributions of terminal relative displacements ($x - c$) on the multi-frequency wave task. Black lines and white dots denote the median and mean, respectively.

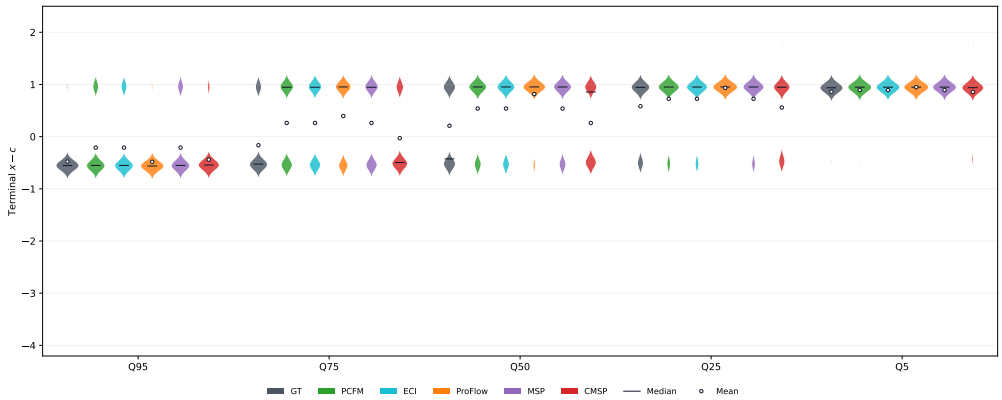


Figure 15: Violin distributions of terminal relative displacements $(x - c)$ on the rational singular curve task. Black lines and white dots denote the median and mean, respectively.

E More Details and Visualization of High-Dimensional Benchmarks

E.1 Constraint Formulation and Physical Hard Constraints

To evaluate zero-shot constraint enforcement across distinct physical mechanisms, we formulate hard physical equality constraints $C(u) = \mathbf{0}$ for three canonical partial differential equation (PDE) benchmarks.

Heat Equation We consider the one-dimensional unsteady heat diffusion process defined on $(x, t) \in \mathcal{X} = [0, X_{\text{max}}] \times [0, T_{\text{max}}]$:

$$\mathcal{P}(u) \equiv \frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} = 0, \tag{91}$$

where $\alpha > 0$ denotes the thermal diffusivity. The physical constraints are constructed by coupling Dirichlet boundary conditions with sparse internal spatiotemporal observations. The constraint vector $C_{\text{Heat}}(u)$ is formulated as:

$$C_{\text{Heat}}(u) = \begin{bmatrix} u(0,t) - g_L(t) \\ u(X_{\text{max}}, t) - g_R(t) \\ \mathcal{H}_{\text{obs}}(u) - y_{\text{obs}} \end{bmatrix} = \mathbf{0}, \tag{92}$$

where $g_L(t)$ and $g_R(t)$ specify the temporal boundary profiles, and $\mathcal{H}_{\text{obs}} : \mathcal{U} \to \mathbb{R}^{m_{\text{obs}}}$ represents a sparse linear masking operator extracting measurements at localized sensor coordinates.

Reaction-Diffusion Equation We study a nonlinear reaction-diffusion system featuring cubic reaction kinetics:

$$\mathcal{P}(u) \equiv \frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} - R(u) = 0, \quad \text{where } R(u) = \gamma u(1 - u^2), \tag{93}$$

with diffusion coefficient D and reaction rate γ. The hard physical constraint combines boundary flux conditions with partial interior spatial field observations, defining the residual operator $C_{\text{RD}}(u)$:

$$C_{\text{RD}}(u) = \begin{bmatrix} \mathcal{B}_{\text{flux}}(u) - q_{\text{bc}} \\ \mathcal{H}_{\text{sparse}}(u) - y_{\text{obs}} \end{bmatrix} = \mathbf{0}, \tag{94}$$

where $\mathcal{B}_{\text{flux}}$ denotes the boundary derivative operator, and $\mathcal{H}_{\text{sparse}}$ extracts localized state values across fixed temporal snapshots.

Burgers' Equation To test robustness under steep gradients and advective shock waves, we examine the viscous Burgers' equation:

$$\mathcal{P}(u) \equiv \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = 0, \tag{95}$$

where $\nu \geq 0$ represents the kinematic viscosity. Hard physical constraints are specified via Dirichlet boundary profiles paired with downstream spatial trajectory measurements:

$$C_{\text{Burgers}}(u) = \begin{bmatrix} u(0,t) - u_{\text{in}}(t) \\ u(X_{\max}, t) - u_{\text{out}}(t) \\ \mathcal{H}_{\text{shock}}(u) - y_{\text{obs}} \end{bmatrix} = \mathbf{0}, \tag{96}$$

where $\mathcal{H}_{\text{shock}}$ measures solution values across downstream observation locations, enforcing physical mass flux consistency and trajectory alignment around moving shock fronts.

E.2 FFM Model Pretraining

To evaluate zero-shot physics-constrained sampling in functional spaces, we employ Functional Flow Matching (FFM) Kerrigan et al. (2023) to pretrain unconstrained generative vector fields v_θ across representative PDE benchmarks. Following standard functional flow modeling protocols Utkarsh et al. (2025), all generative backbones are trained purely on unconstrained solution ensembles, with physical and observational constraints enforced strictly post-hoc during inference.

Dataset Ensembles and Base Prior Measures We construct offline training datasets of ground-truth spatiotemporal PDE trajectories by systematically varying initial conditions, boundary fluxes, or physical system parameters:

- **Heat Equation:** The dataset comprises $5{,}000$ analytical spatiotemporal trajectories generated on a 100×100 uniform grid ($(x,t) \in [0, 2\pi]\times$) by sampling diffusion coefficients $\alpha \sim \mathcal{U}(1,5)$ and initial phase shifts $\phi \sim \mathcal{U}(0,\pi)$.
- **Reaction-Diffusion Equation:** The dataset contains $6{,}400$ numerical solutions solved on a 128×100 space-time grid via a semi-implicit finite difference scheme, constructed from 80 randomized initial profile combinations and 80 boundary flux configurations.
- **Viscous/Inviscid Burgers' Equation:** The dataset consists of $6{,}400$ numerical solutions discretized on a 101×101 grid computed via a Godunov finite volume solver, pairing 80 randomized step-like initial profile centers $p_{\text{loc}} \sim \mathcal{U}(0.2, 0.8)$ with 80 Dirichlet boundary conditions $u_{\text{bc}} \sim \mathcal{U}(0,1)$.

Across all experiments, the base noise prior μ_0 is uniformly modeled as a Matérn Gaussian Process (GP) with a smoothness parameter $\nu = 0.5$, a correlation length scale $l = 0.001$, and marginal variance $\sigma^2 = 1.0$, implemented via GPyTorch.

Neural Operator Backbone Configurations Across all benchmarks, the time-dependent velocity field $v_\theta(u_\tau, \tau)$ is parameterized by a discretization-invariant Fourier Neural Operator (FNO) Li et al. (2021). The network input concatenates the spatial state u_τ, spatial grid coordinates, and sinusoidal temporal Fourier embeddings. The backbone consists of 4 Fourier layers with 32 Fourier modes truncated along spatiotemporal frequency dimensions, 64 hidden channels (width), and 256 projection channels.

Optimization Protocols and Hyperparameters All operator backbones are trained by minimizing the L^2 functional velocity matching loss $\mathcal{L}_{\text{FFM}}(\theta)$ using the Adam optimizer with an initial learning rate $\eta_0 = 3 \times 10^{-4}$, momentum parameters $(\beta_1, \beta_2) = (0.9, 0.999)$, and zero weight decay. A ReduceLROnPlateau scheduler reduces the learning rate by a factor of 0.5 whenever the validation loss plateaus for 10 consecutive validation checks, down to a minimum threshold of 1×10^{-4}.

Across all experimental setups, the batch size is set to 256, and all operator models are trained for $20{,}000$ optimization steps. Once pretraining completes, all network parameters θ are permanently frozen during zero-shot inference.

E.3 Sampling Configurations and Hyperparameters

To ensure standardized and equitable comparisons across all zero-shot inference experiments, all evaluated generative samplers are executed under identical temporal discretization budgets and numerical solver configurations. Across all benchmarks, the generative flow trajectories are integrated using a uniform Euler ODE solver with $N = 100$ discretization steps ($\Delta t = 0.01$). For each method, we generate an ensemble of $N_{\text{sample}} = 512$ independent trajectories per task to evaluate all statistical moments, distributional distances, and physical feasibility metrics.

Regarding operator hyperparameters and solver configurations, the inner correction loop for ECI (Linear & Nonlinear) is set to 3 iterations per timestep for non-overlapping root-finding and constraint solving. For nonlinear physical constraints, the terminal manifold closure operator across PCFM, MSP, ProFlow, and our CMSP employs a single Gauss–Newton iteration ($N_{\text{Newton}} = 1$), which is sufficient to project candidate predictions onto the zero-residual manifold $\mathcal{M}_1 = \{u \mid C(u) = 0\}$. Additionally, for our CMSP, the normal-space energy dissipation via Pre-Image Invariant Spectral Transfer (PIS) executes with $K = 1$ descent iteration per timestep, exploiting the dominant Krylov subspace direction for linear transfer without incurring unnecessary runtime overhead.

To systematically evaluate zero-shot constraint enforcement under diverse physical configurations and parameter shifts , we detail the dataset construction protocols and constraint selection choices for each PDE benchmark below:

- **Heat Equation:** For the Heat Equation benchmark, we construct the evaluation dataset using five evenly spaced phase parameter configurations $\theta \in \{\frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}\}$, where $\theta = \frac{5\pi}{4}$ represents an Out-of-Distribution (OOD) scenario and the remaining four correspond to in-distribution settings. For our quantitative evaluation, we randomly select $\theta = \pi$ as the target constraint parameter.
- **Reaction-Diffusion Equation (IC):** For the Reaction-Diffusion benchmark, we construct the evaluation dataset using five deterministic constant initial condition (IC) profiles $u_0(x) \equiv u_{\text{ic}} \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$, each paired with 512 randomized boundary condition (BC) flux pairs (g_L, g_R) sampled within training limits. Here, $u_{\text{ic}} = 1.2$ represents an Out-of-Distribution (OOD) scenario. For our quantitative evaluation, we randomly select $u_{\text{ic}} = 0.6$ as the target initial condition constraint evaluated across all 512 boundary condition test instances.
- **Burgers' Equation (IC):** For the initial condition evaluation in the Burgers' Equation benchmark, we construct five step-like profile centers $p_{\text{loc}} \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$, where $p_{\text{loc}} = 1.0$ serves as the OOD configuration. For our quantitative evaluation, we randomly select $p_{\text{loc}} = 0.6$ as the target initial condition constraint for quantitative testing.
- **Burgers' Equation (BC):** For the boundary condition evaluation in the Burgers' Equation benchmark, we construct five deterministic Dirichlet boundary profiles $u_{\text{bc}} \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$, where $u_{\text{bc}} = 1.2$ serves as the OOD configuration. For our quantitative evaluation, we randomly select $u_{\text{bc}} = 0.0$ as the target boundary condition constraint for quantitative testing.

E.4 More Visualizations

We plot the mean and standard deviation solution profiles for the Ground Truth (GT) and the evaluated models; visual comparisons demonstrate that our model consistently achieves superior solution similarity compared to competing baselines. (See Figs. 16–18.)

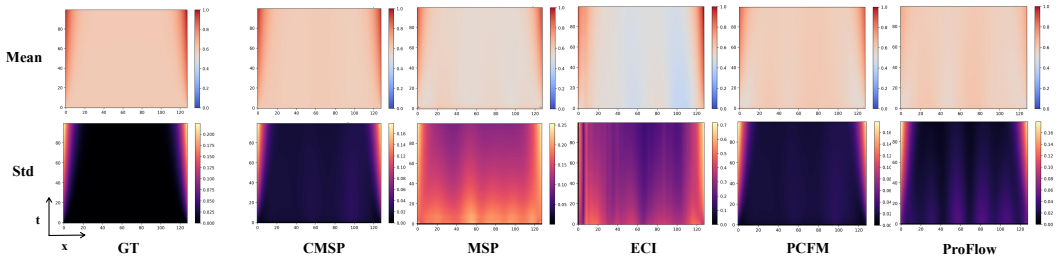


Figure 16: Comparison of generated solutions on the Reaction-Diffusion IC task. CMSP achieves the closest agreement with the ground-truth (GT) reference posterior.

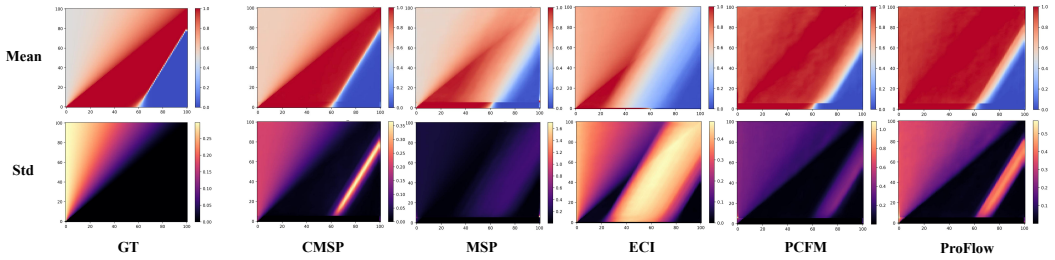


Figure 17: Comparison of generated solutions on the Burgers IC task. CMSP achieves the closest agreement with the ground-truth (GT) reference posterior.

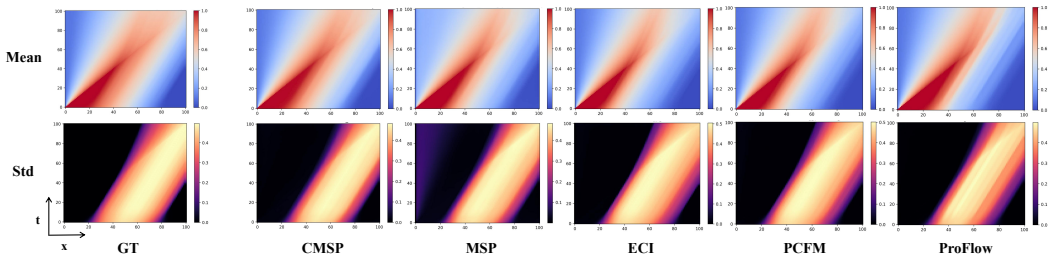


Figure 18: Comparison of generated solutions on the Burgers BC task. CMSP achieves the closest agreement with the ground-truth (GT) reference posterior.

F Further Experiments

F.1 High-Dimensional Phase-Space Posterior Sweep

While Section D validates the measure-preservation capability of CMSP on diverse two-dimensional geometric topologies, a critical open question is whether this superiority persists in discretized infinite-dimensional functional spaces governed by partial differential equations. To investigate this, we follow the experimental benchmark introduced in MSP (Anonymous, 2026). Here, the configurations and hyperparameters of all generative models and baselines are strictly identical to those employed in the high-dimensional heat experiment in Section 5.3.

Table 7 presents the comprehensive quantitative results across all evaluated phase configurations, reporting the Mean Field MSE (MMSE), Standard Deviation Field MSE (SMSE), and the Fréchet Poseidon Distance (FPD).

The results demonstrate the consistent superiority of CMSP across all five phase configurations. CMSP achieves the lowest error on all 15 metric. Relative to the best non-CMSP baseline in each configuration, CMSP reduces these errors by an average of 77.1%, 58.9%, and 72.1%, respectively. Notably, this advantage persists in the out-of-distribution setting $\phi = 5\pi/4$. These results demonstrate that CMSP maintains accurate posterior moments and distributional fidelity across varying phase configurations, including an unseen phase outside the prescribed range.

G Limitation

The transition from the exact conditional velocity in Section 4.1 to its tractable approximation in Section 4.2 relies on a local Gaussian approximation of the endpoint conditional distribution and a first-order linearization of the nonlinear constraint manifold. The resulting velocity is therefore an approximate, covariance-weighted conditional velocity that neglects higher order geometric information, including the curvature of the constraint manifold and non-Gaussian features of the endpoint distribution. In strongly nonlinear regimes, where such effects become significant over the region of posterior mass, the approximation may misestimate the conditional mean and consequently the velocity field. Although terminal projection can still enforce constraint feasibility, it does not

Table 7: Quantitative benchmark results on the Linear Heat Inverse Problem across varying initial phase configurations $\phi \in [\pi/4, 5\pi/4]$. The metrics evaluate the fidelity of the inferred posterior: Mean Field MSE (MMSE), Standard Deviation Field MSE (SMSE), and Fréchet Poseidon Distance (FPD). Lower (↓) is better. **Bold** indicates the best performance.

Metric	CMSP (Ours)	PCFM	MSP	ECI	ProFlow
Initial Phase $\phi = \pi/4$ ***(***$45°$***)***					
MMSE ($\times 10^{-2}$)	$\mathbf{0.012}$	2.073	0.407	6.551	3.495
SMSE ($\times 10^{-2}$)	$\mathbf{0.026}$	1.218	0.376	7.410	0.081
FPD	$\mathbf{0.959}$	8.130	6.730	12.783	9.465
Initial Phase $\phi = \pi/2$ ***(***$90°$***)***					
MMSE ($\times 10^{-2}$)	$\mathbf{0.057}$	2.748	0.115	10.378	5.199
SMSE ($\times 10^{-2}$)	$\mathbf{0.043}$	1.299	0.387	12.984	0.103
FPD	$\mathbf{1.590}$	7.363	6.264	11.723	11.450
Initial Phase $\phi = 3\pi/4$ ***(***$135°$***)***					
MMSE ($\times 10^{-2}$)	$\mathbf{0.024}$	1.889	0.608	8.342	2.377
SMSE ($\times 10^{-2}$)	$\mathbf{0.025}$	1.453	0.399	12.294	0.040
FPD	$\mathbf{1.104}$	7.446	7.518	12.655	8.149
Initial Phase $\phi = \pi$ ***(***$180°$***)***					
MMSE ($\times 10^{-2}$)	$\mathbf{0.008}$	1.262	1.989	8.129	0.709
SMSE ($\times 10^{-2}$)	$\mathbf{0.009}$	1.499	0.574	9.969	0.016
FPD	$\mathbf{0.679}$	6.476	10.213	12.084	4.409
Initial Phase $\phi = 5\pi/4$ ***(***$225°$***, Out-of-Distribution)***					
MMSE ($\times 10^{-2}$)	$\mathbf{0.170}$	0.506	3.875	4.546	0.298
SMSE ($\times 10^{-2}$)	$\mathbf{0.008}$	1.493	1.204	6.718	0.061
FPD	$\mathbf{2.123}$	5.946	12.893	11.099	3.042

recover the second-order information omitted from the velocity approximation, potentially reducing the fidelity of the resulting posterior samples.